



Introduction to Computational Text Analysis

Introduction to Corpus Linguistics

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What is Computational Text Analysis?

Computational text analysis is the use of algorithms to analyze texts.

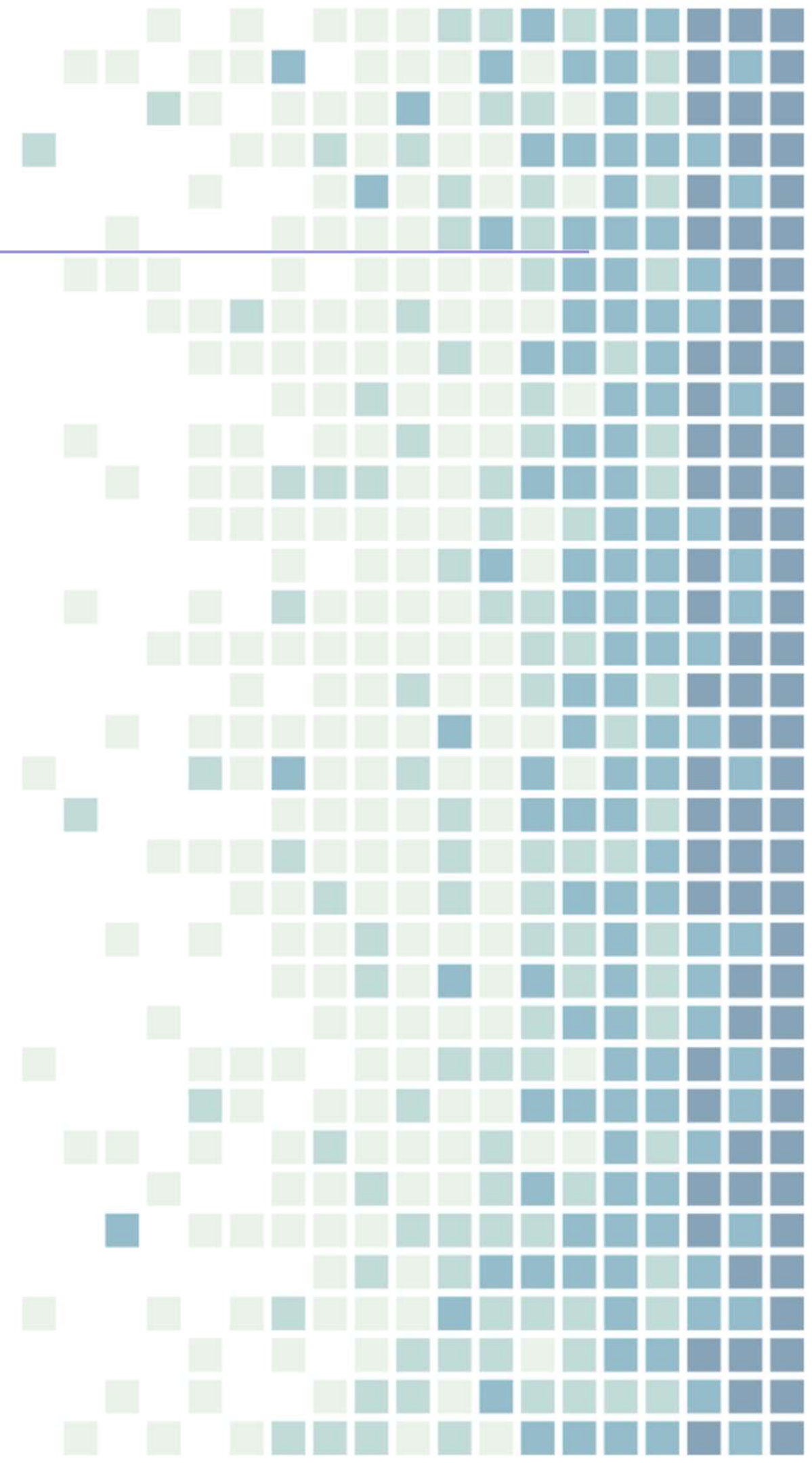
It's data analysis for written language.

It combines ideas from:

- Linguistics,
- Computer Science,
- Statistics & Data Analysis

but also:

- Digital Humanities
- Literary Studies,
- Cognitive Science
- and also others, depending on the research question



What is Computational Text Analysis?

Computational Text Analysis is not an end in itself.

It is a tool to extract information, check hypothesis and find new insights from any textual information.



What is Computational Text Analysis?

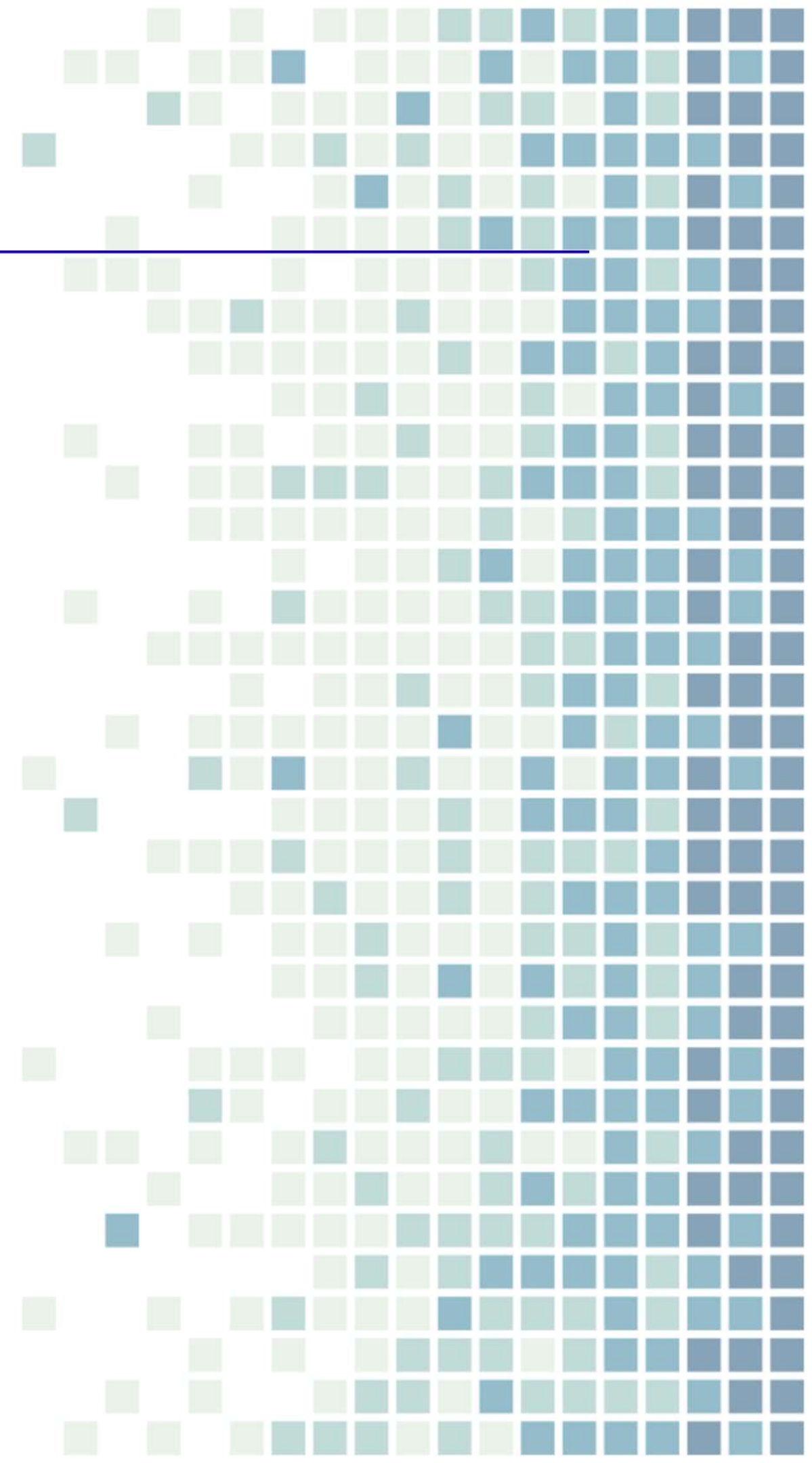
Computational text analysis is inherently interdisciplinary and can should support research in any field involving textual material.

It is a **SUPPORTING DISCIPLINE** for any field that works with text.

What is Computational Text Analysis?

This does not replace expert knowledge.

HUMAN EXPERTISE remains essential.



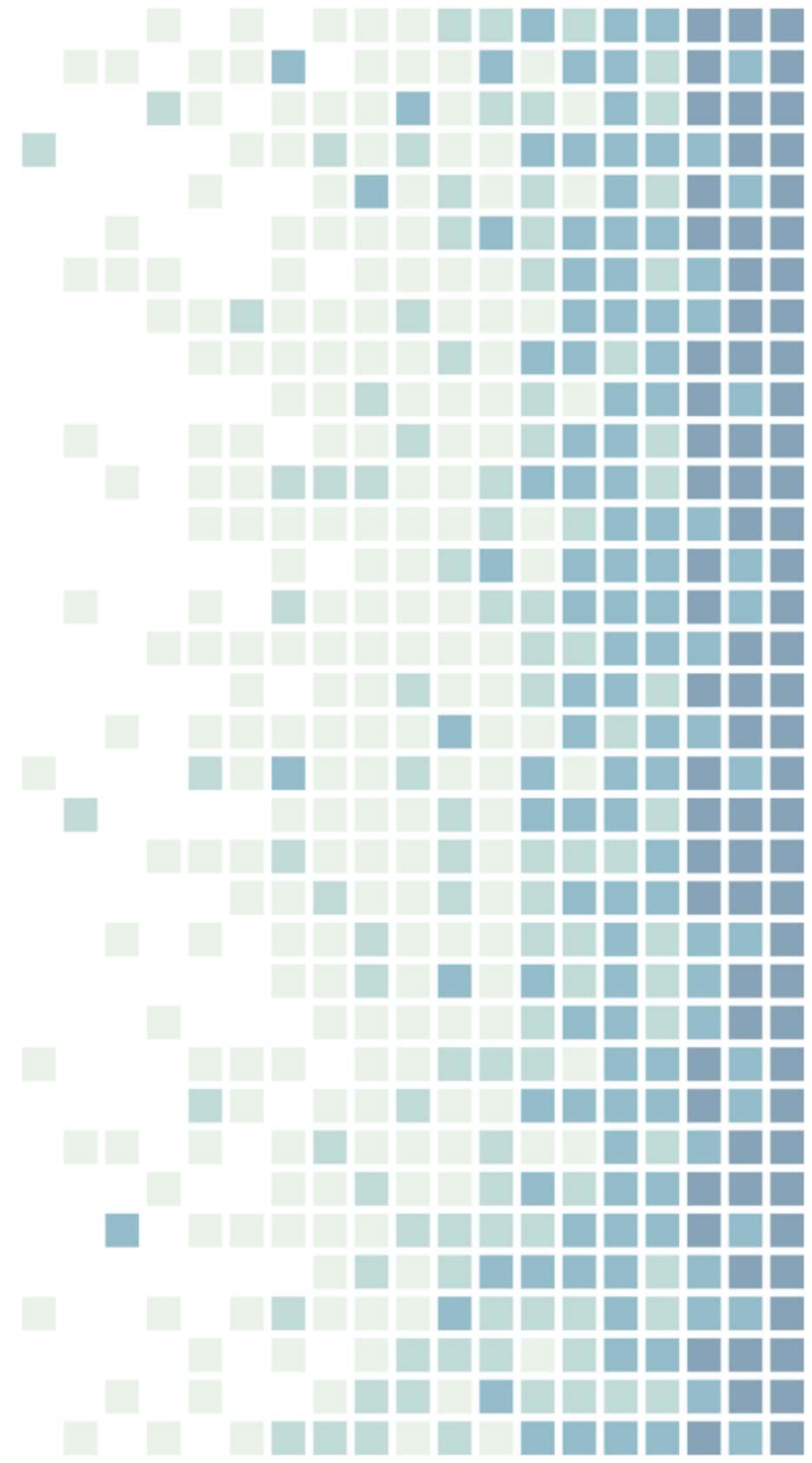
What is Computational Text Analysis?

- It complements than replace domain expertise (we still need to think).
- A big part of data science is in interpreting our results.
- Domain knowledge must be incorporated to get the best of Digital Humanities.

What is Computational Text Analysis?

- Research in the vein of Digital Humanities can give different answers from those we are used to or from those we know from „classical research”.
- However, it is necessary to return from graphs, charts, plots, and AI to the texts and to analyse and interpret them in the face of these data.

This leads us to Corpus Linguistics...



Corpus linguistics

What's a corpus?

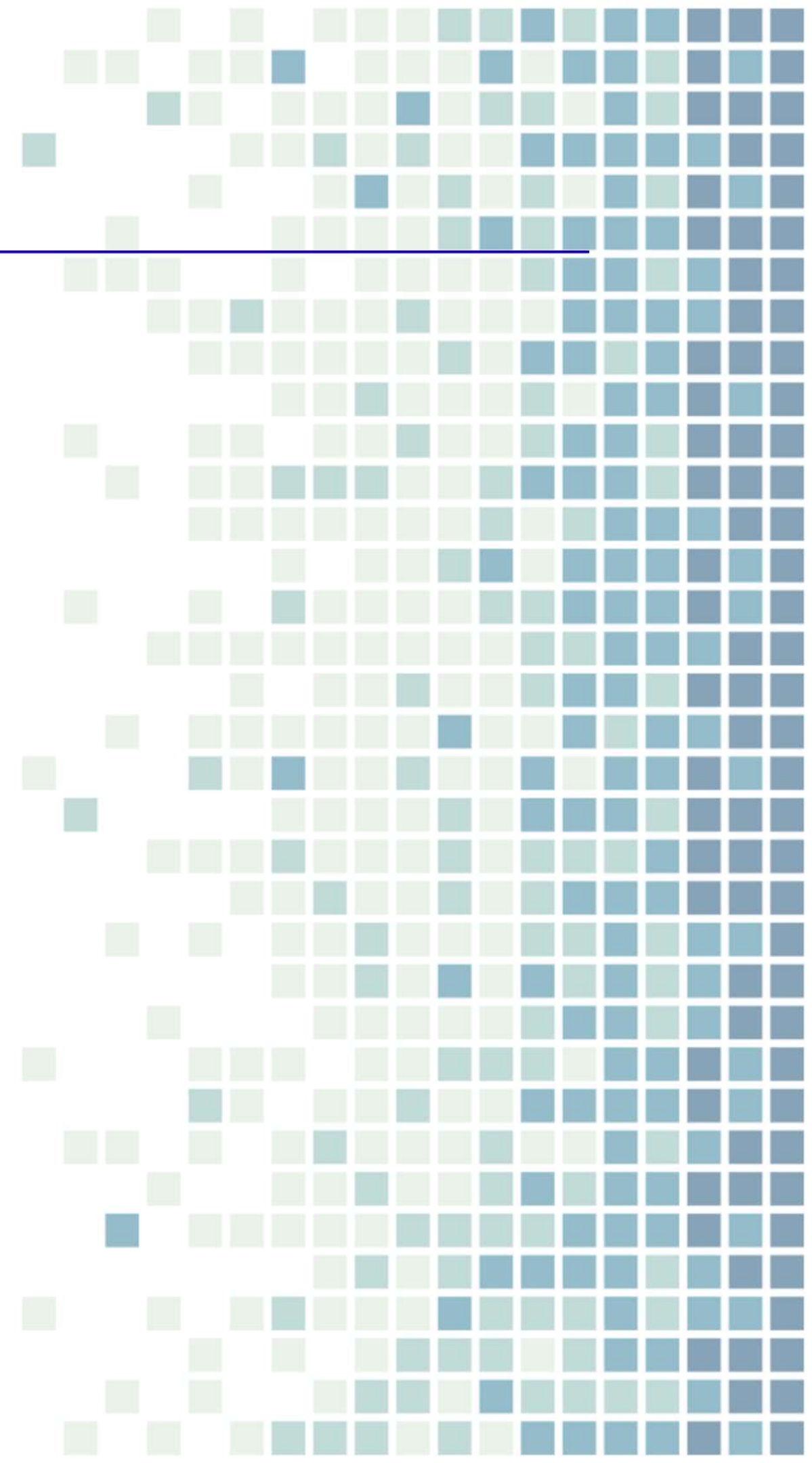
A collection of texts selected according to specific criteria and of sufficient size to support linguistic research.



Corpus linguistics

Additionally:

- digital form
- searchable
- additional layers of information



Research potential

- a source of objective knowledge about language
- a source of linguistic data (for testing theories, illustrating hypotheses, etc.)
- helps ensure the objectivity of research
- enables the replication of research
- provides access to authentic language use
- results are derived from large amounts of data that would otherwise be impossible to process manually

But also...

- Since a linguistic corpus is a large collection of texts and/or spoken utterances, it can be assumed to represent — to some extent — the language and its patterns of use;
- but also as part of the **collective knowledge** and **stereotypes** embedded in it.



An empirical approach

The Corpus Approach is **EMPIRICAL**, analyzing the actual patterns of language use in natural texts



To sum up: Purposes of Corpus linguistics

Switch from subjective analysis to **OBJECTIVE, REPEATABLE** and **MEASURABLE** results. We no longer rely on our intuition or convictions, but on **DATA**.

Real-world example

Can I use the word oftentimes in a scientific paper?



Real-world example

Can I use the word oftentimes in a scientific paper?

We can rely on our intuition, but instead of that we can try to measure the popularity of this word:

<https://books.google.com/ngrams/>

Google Books Ngram Viewer

🔍

oftentimes

✕

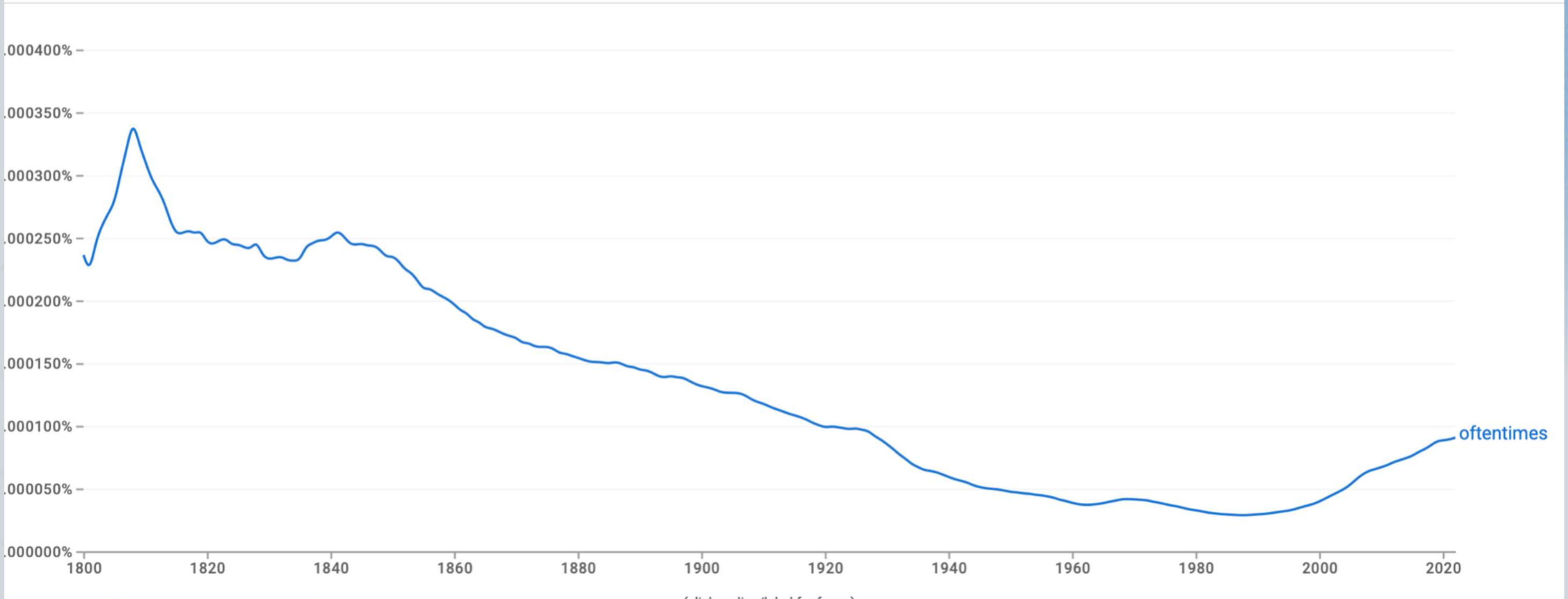
?

1800 - 2022 ▾

English ▾

Case-Insensitive

Smoothing ▾



Real-world example

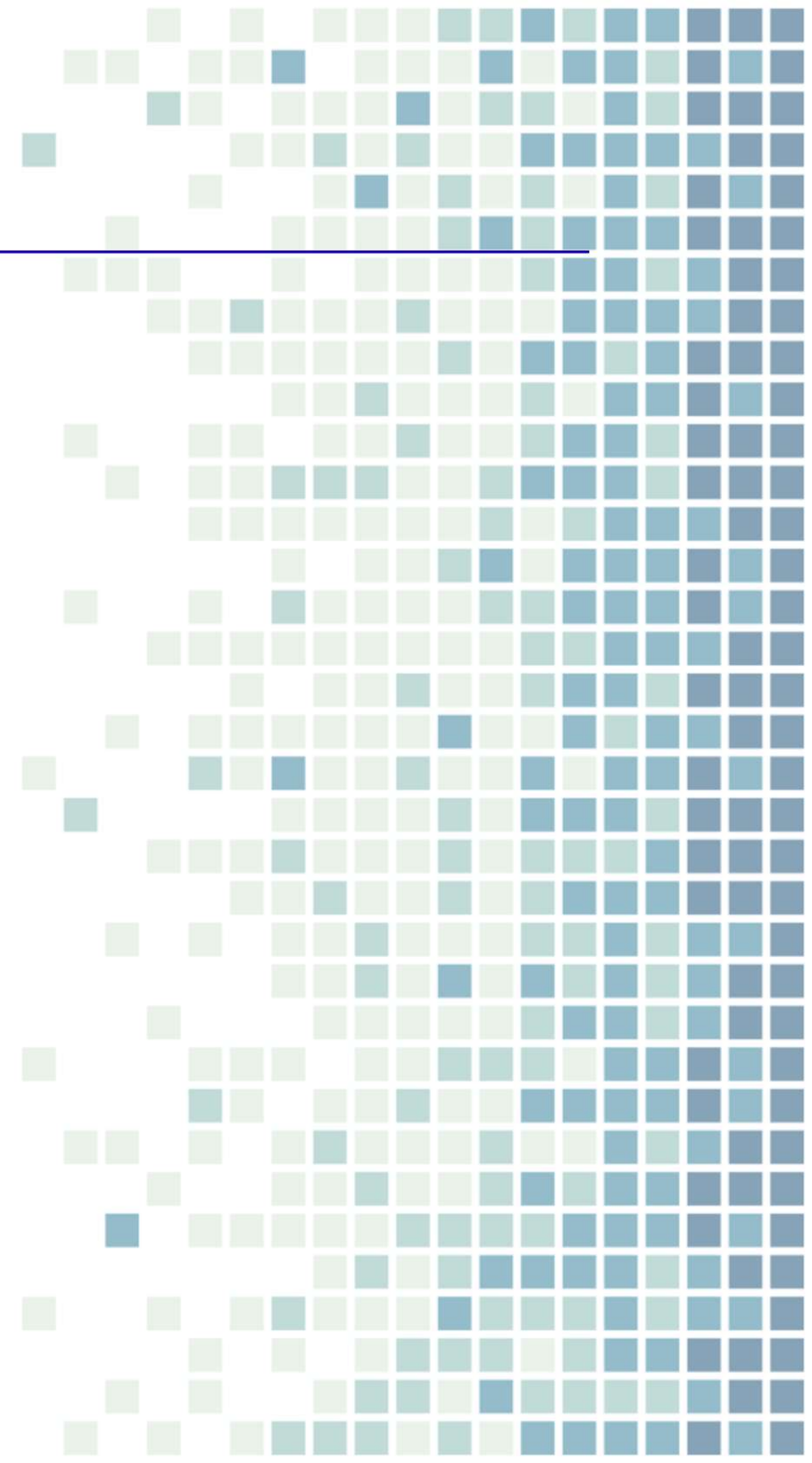
Can I use the word oftentimes in a scientific paper?

Full-fledged corpus search:

<https://www.english-corpora.org/coca/>

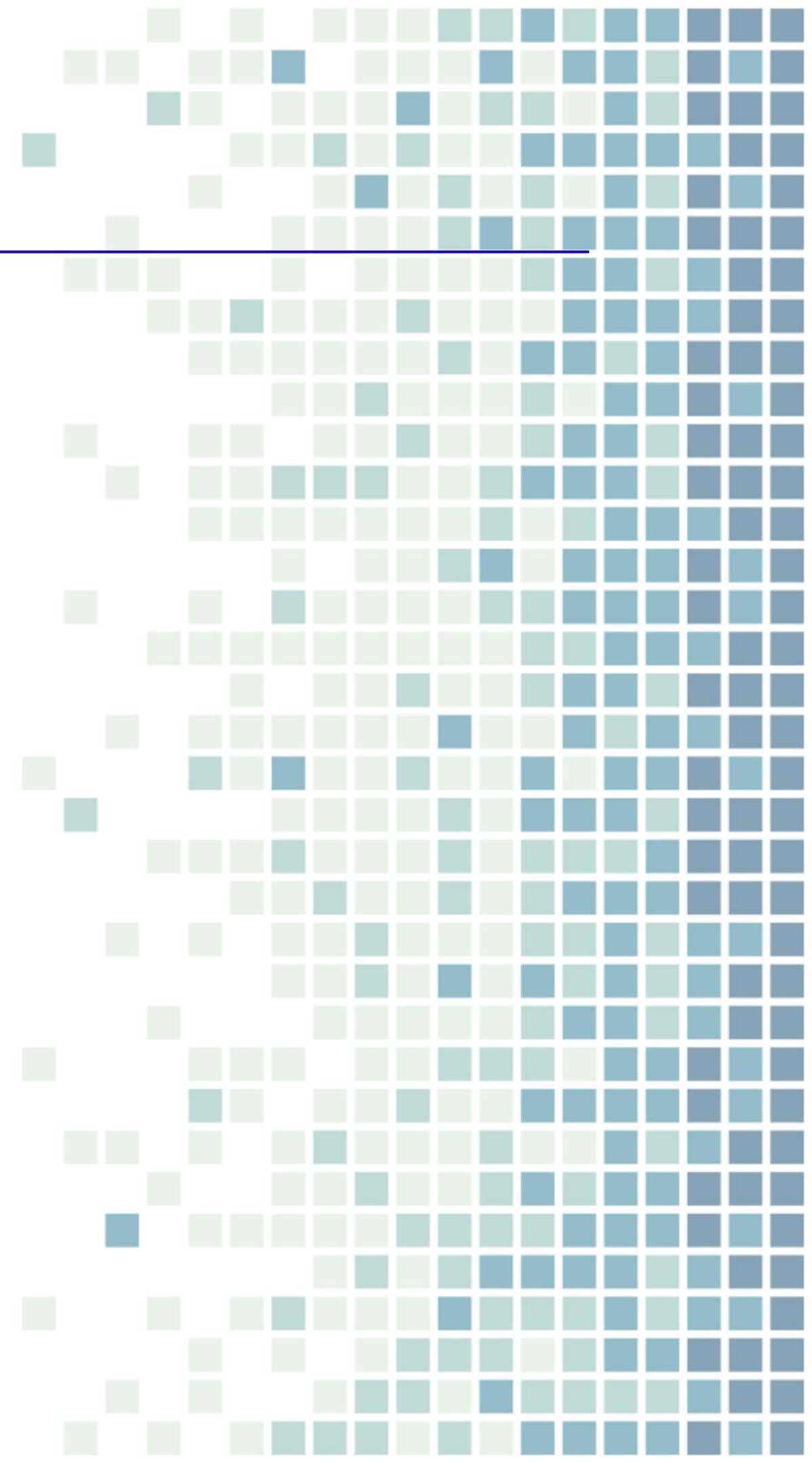
The Paradigm

- quantitative study (distant reading)
- qualitative study (close reading)
- corpus-based
- corpus-driven



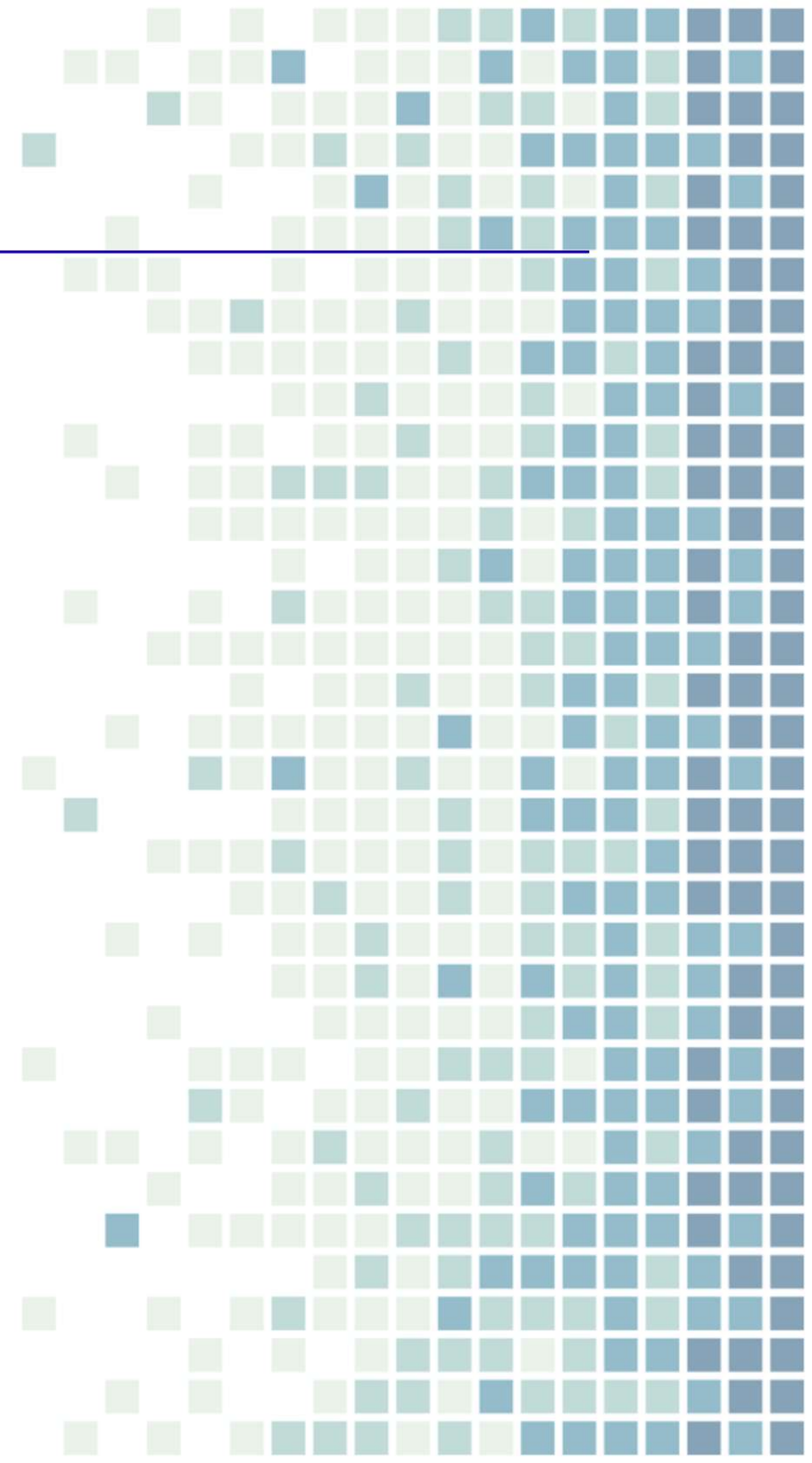
Critique of the Paradigm

- language competence vs performance
- prone to errors
- negative errors (gaps in the data)
- finite data set vs. infinite nature of language



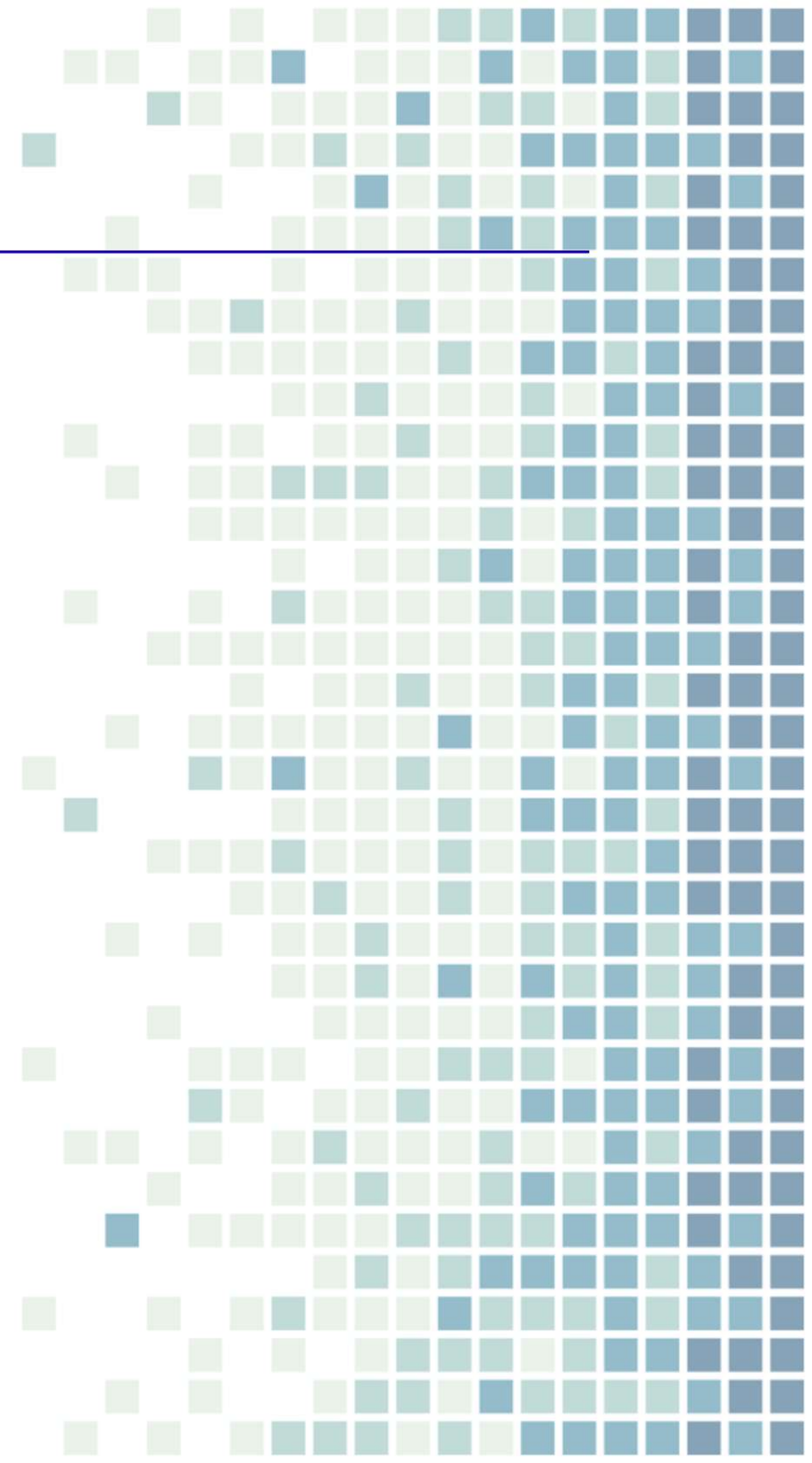
Typology

- reference / opportunistic
- closed / monitor
- monolingual / multilingual
 - parallel / comparable
- written / spoken
- contemporary / historical



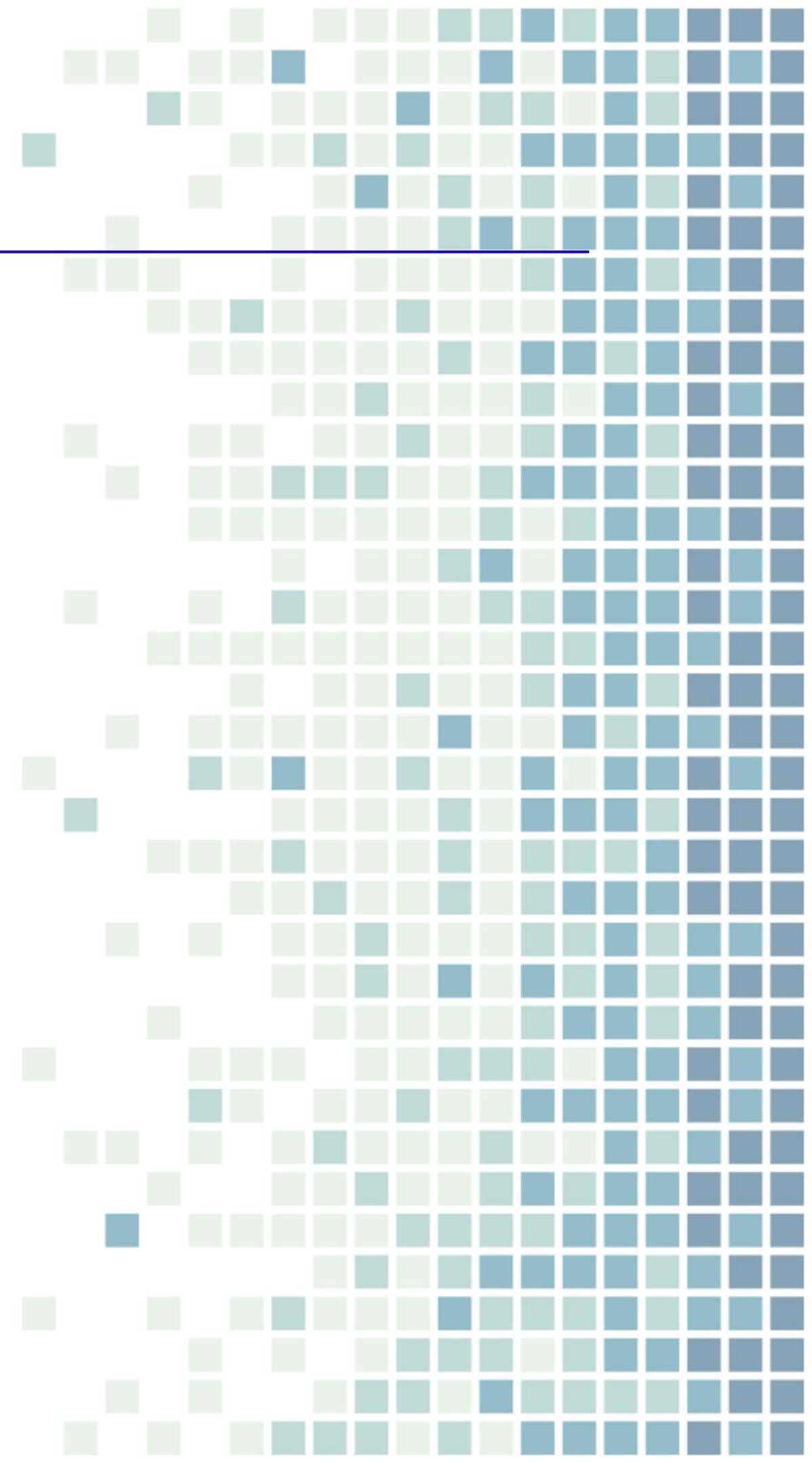
Basic terms

- concordance
- match
- context
- Key Word in Context (KWIC)
- query language
- tagset



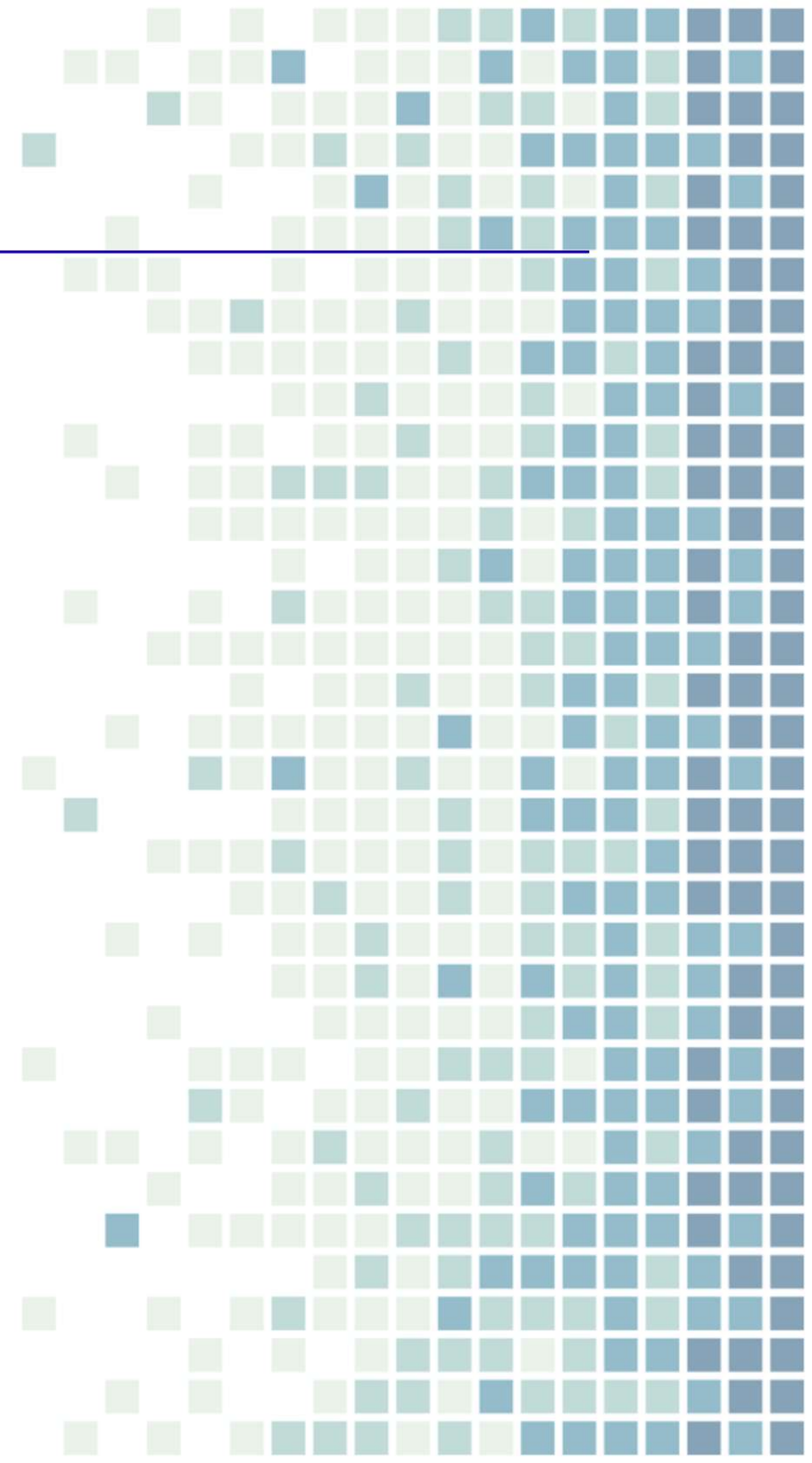
Basic terms

- document
- sample
- type / token
- tokenization
- lemmatization & tagging
- annotation



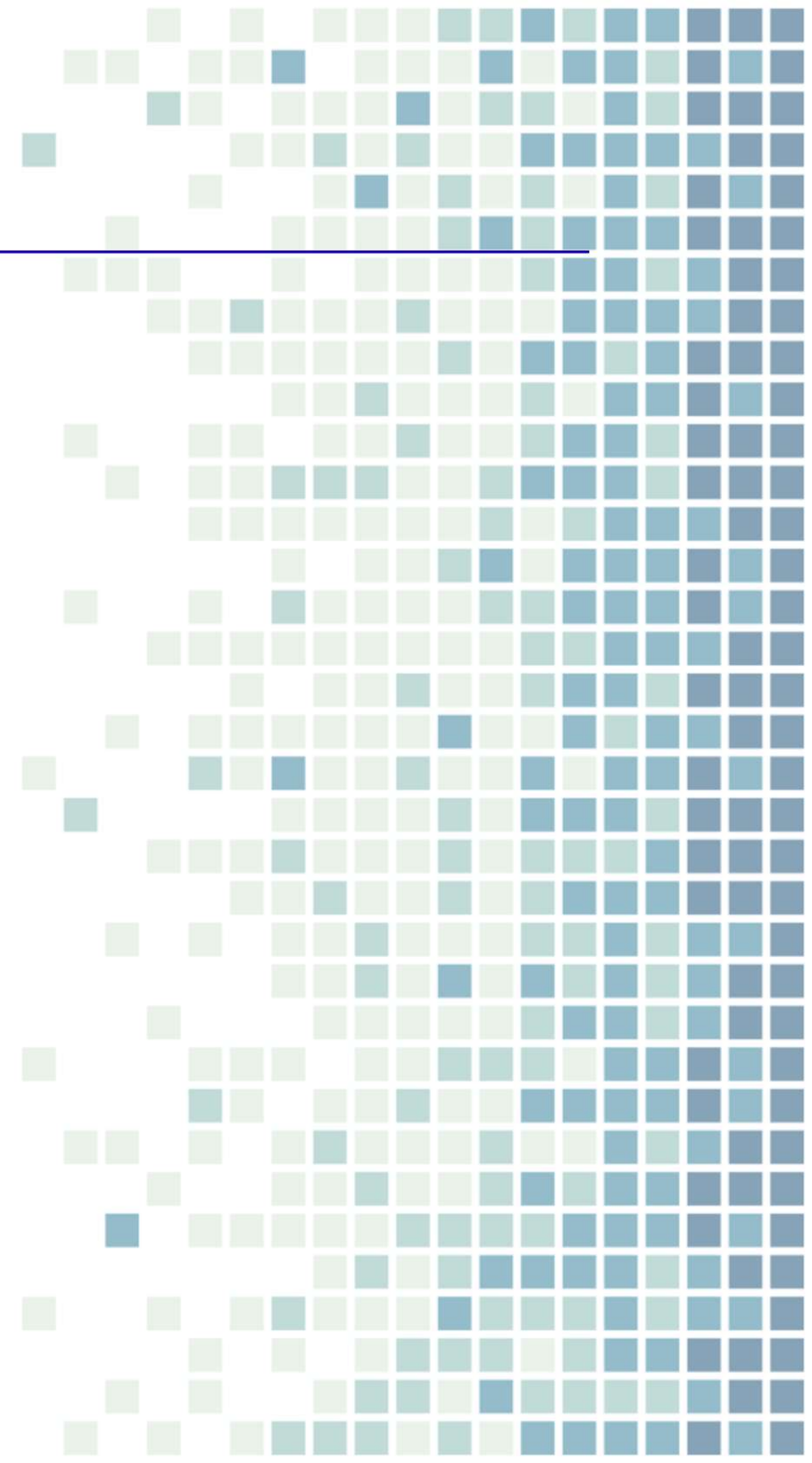
Basic terms: annotation

- metadata
 - title, author, date, source, language, etc...
- structural
 - sentences/phrases/utterances
 - paragraphs, chapters
 - Multiword Units / Expressions
 - ...
- positional
 - lemma, POS, orthographic, ...



Let's go practical: building your own corpus

- AntConc
- VoyantTools
- (No-)SketchEngine
- CWB / CQPWeb
- Many more



Frequency

What does it provide?

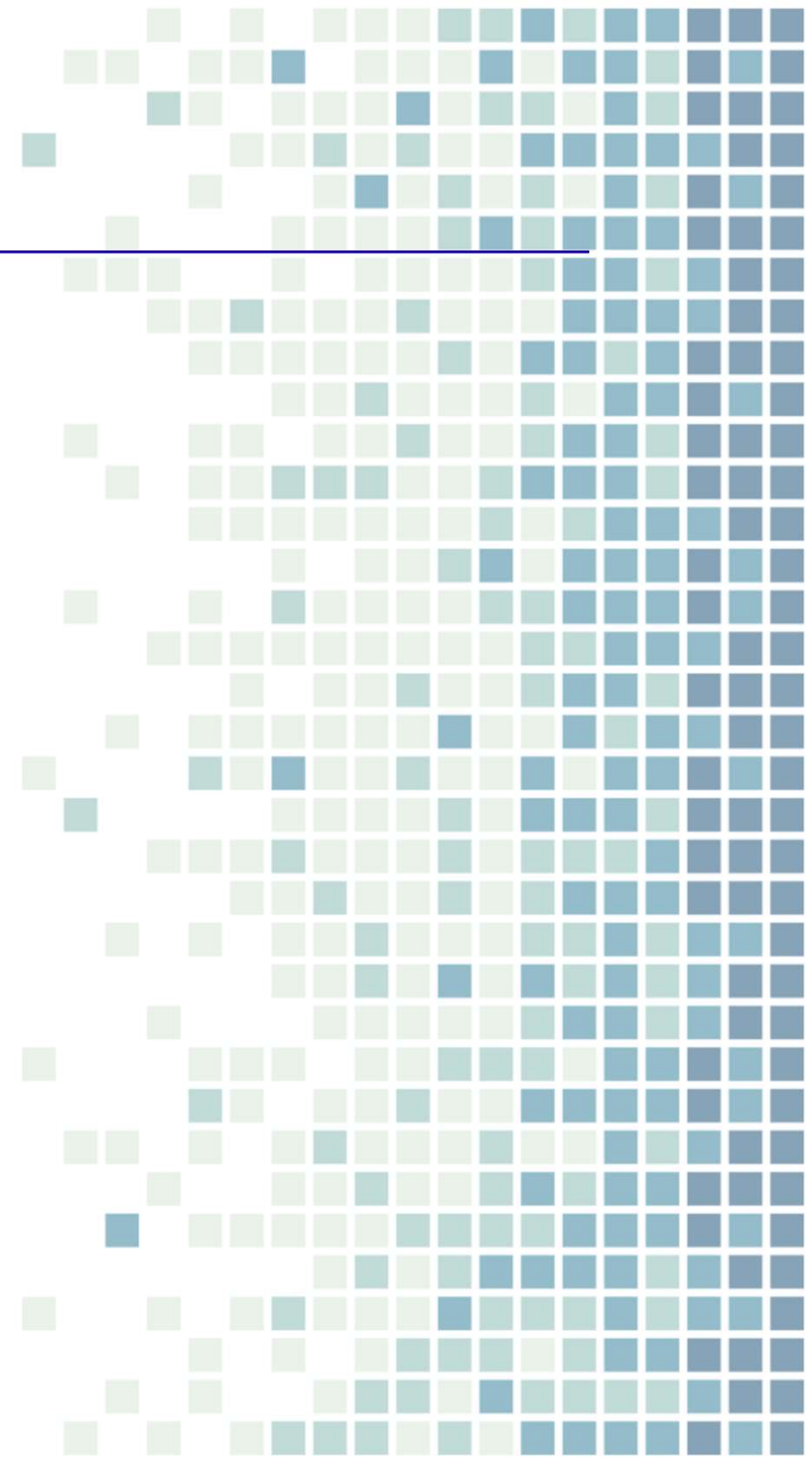
- a bird's-eye view
- an objective, comparable measure
- statistical characteristics

What should you watch out for?

- ignores context
- ignores order
- sensitive to small differences
- strongly dependent on corpus size

Frequency

Example: *dog*

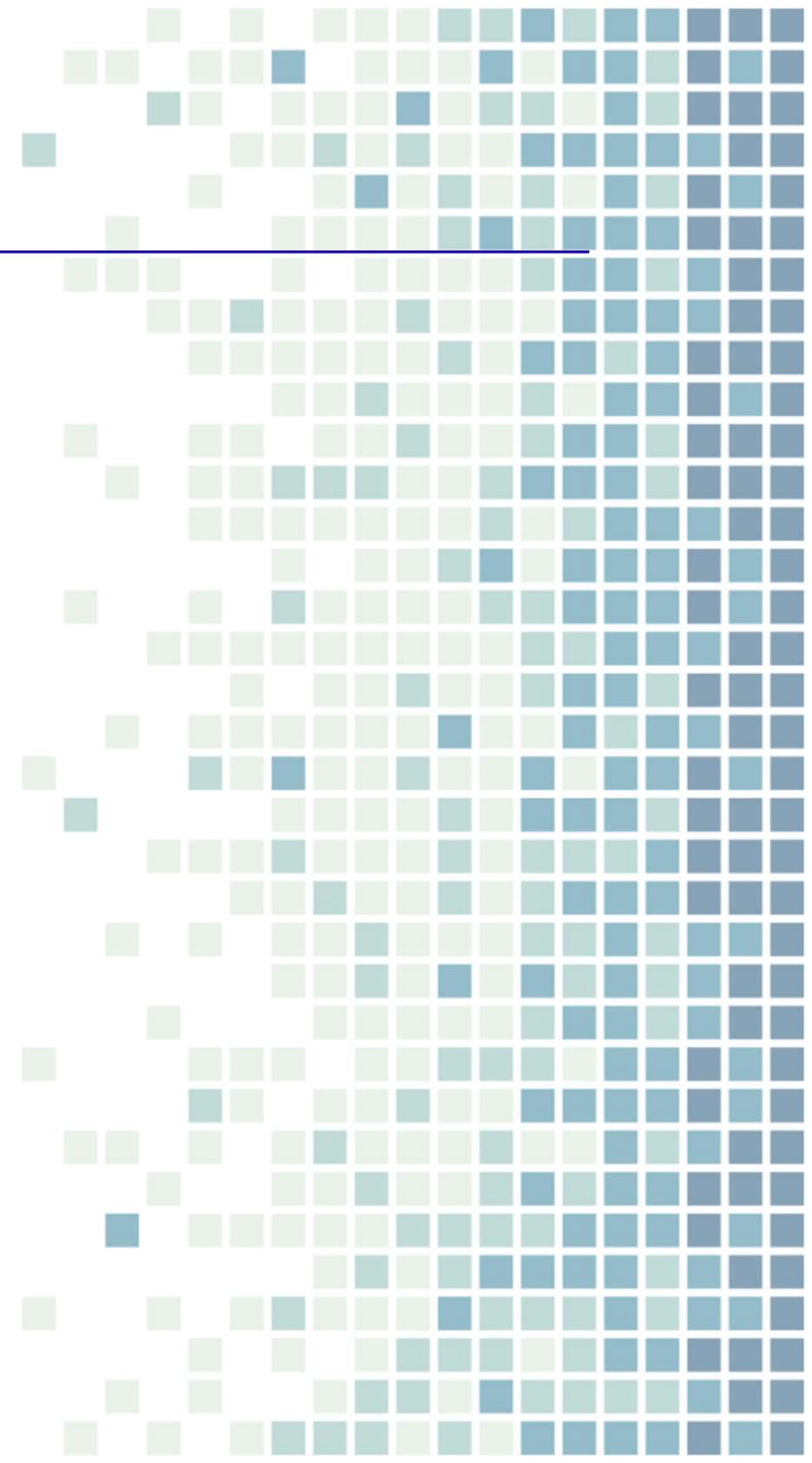


Frequency

Example: *dog*

- 100 English Novels: 1715 hits

Is that a lot or a little?



Frequency

Example: *dog*

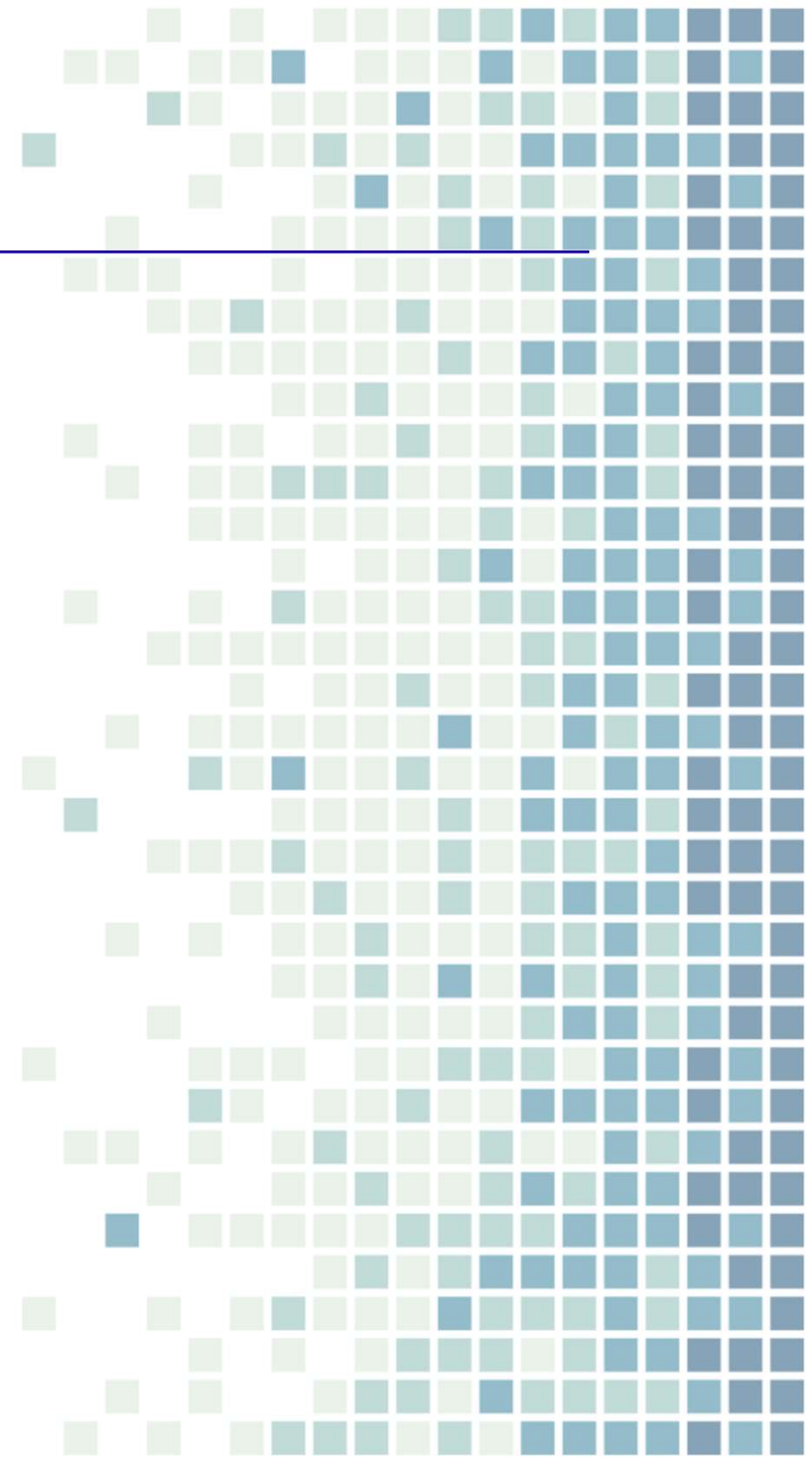
- 100 English Novels: 1715 hits
- BNC: 7780 hits
- COCA: 98847 hits

Does it mean “dog” is 6 times more popular in the BNC and 57 times more popular in the COCA?

Frequency

Corpus Size

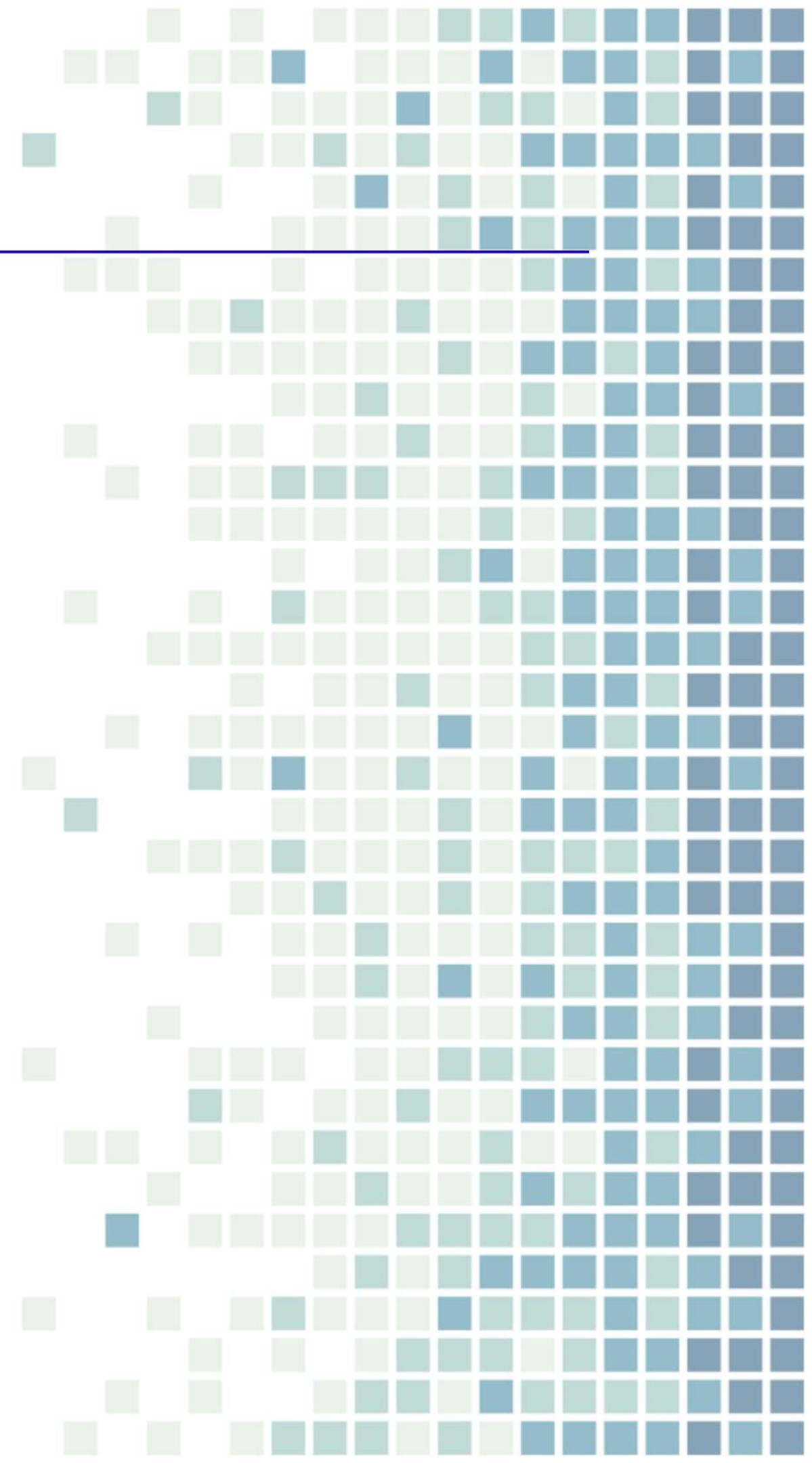
- 100 English Novels: 14 million words
- BNC: ~100 million words
- COCA: ~ 1 billion words number of words



Frequency

Relative Frequency

$$\text{relative frequency (word)} = \frac{\text{frequency (word)}}{\text{total number of words}}$$

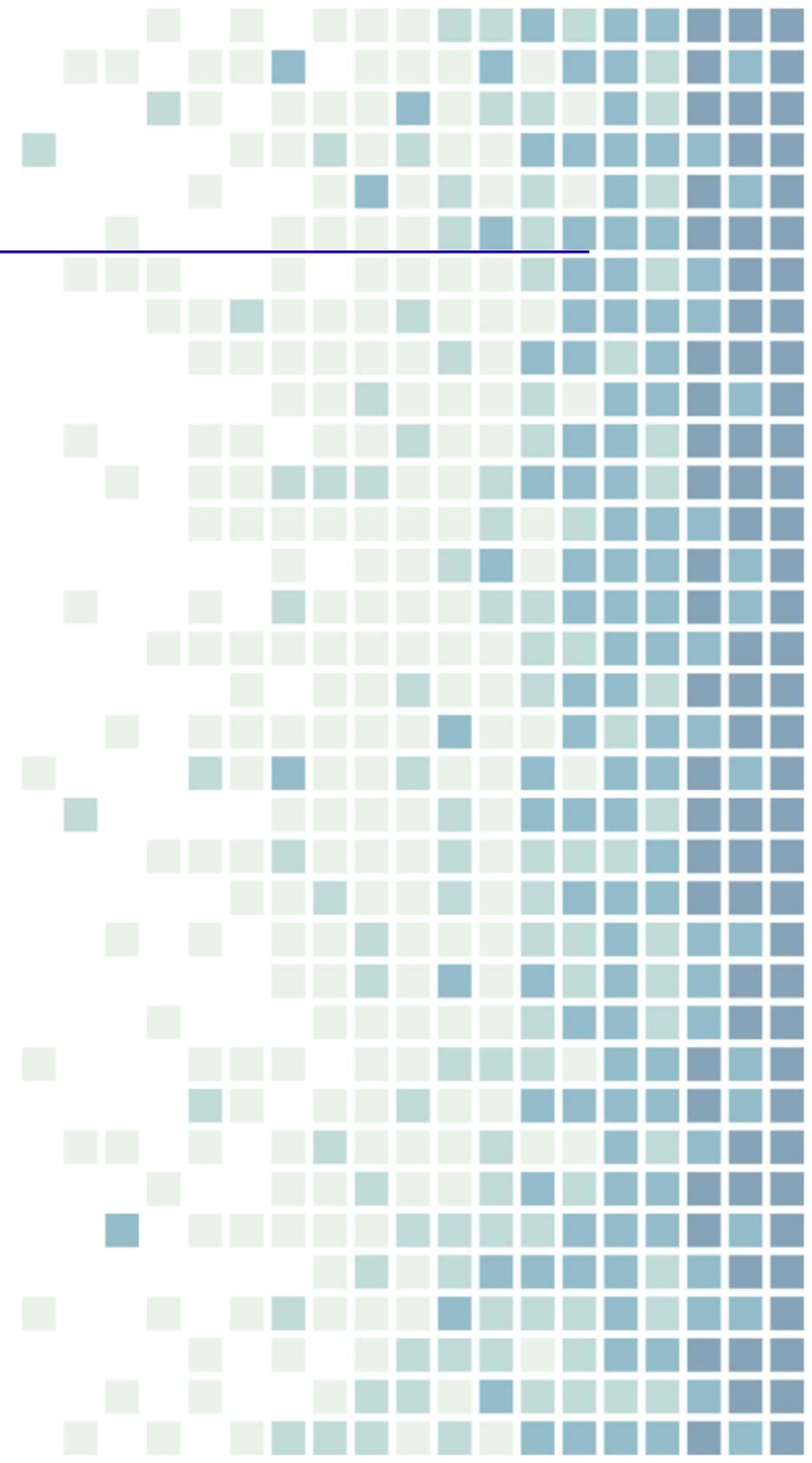


Frequency

dog: relative frequency

- 100 English Novels: 0.00012
- BNC: 0.000078
- COCA: 0.000099

It's actually the other way around



Frequency

dog: relative frequency

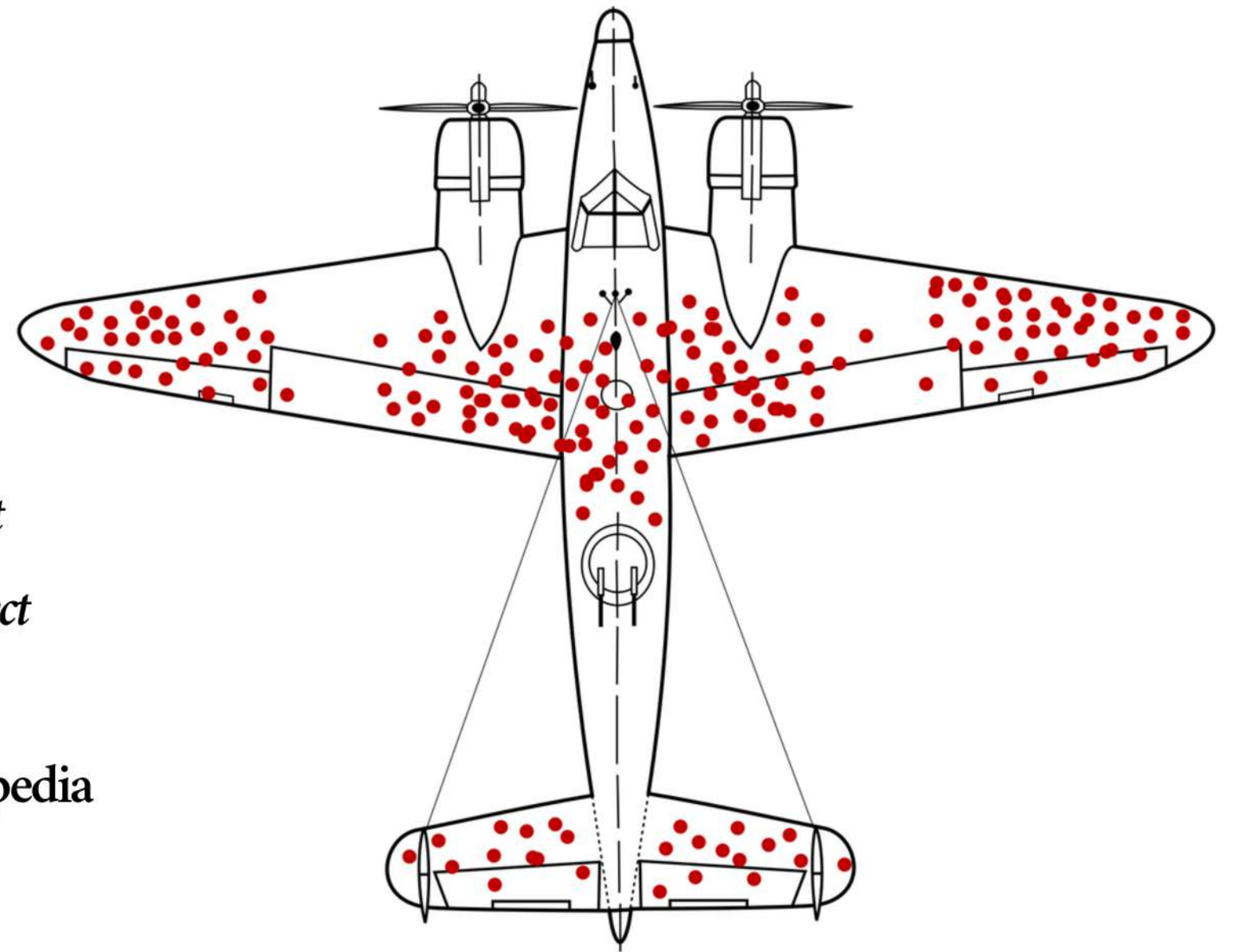
- 100 English Novels: 119.5 words per MILLION
- BNC: 77.8 words per MILLION
- COCA: 98.8 words per MILLION

About gaps in data: Survivorship Bias

https://en.wikipedia.org/wiki/Survivorship_bias

“Survivorship bias is a statistical error that results from concentrating on entities that passed a selection process while overlooking those that did not. This can lead to incorrect conclusions because of incomplete data.”

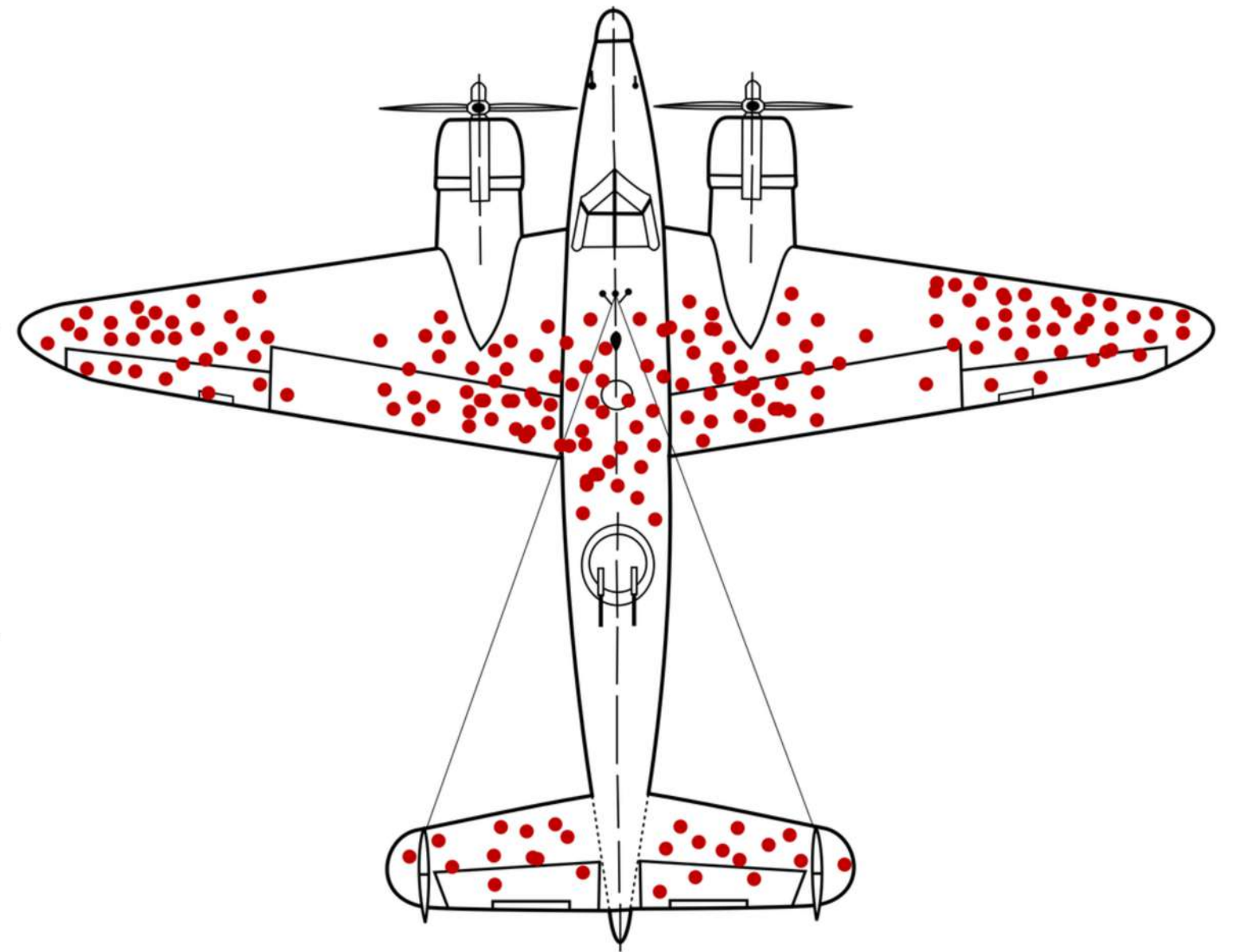
source: Wikipedia



source: Wikipedia

About gaps in data: Survivorship Bias

- During WWII engineers studied returning aircraft to decide where additional armor was needed.
- They analyzed which parts of the planes had been hit most often during combat.
- At first, the conclusion seemed obvious: reinforce the areas with the most bullet holes.



source: Wikipedia

About gaps in data: Survivorship Bias

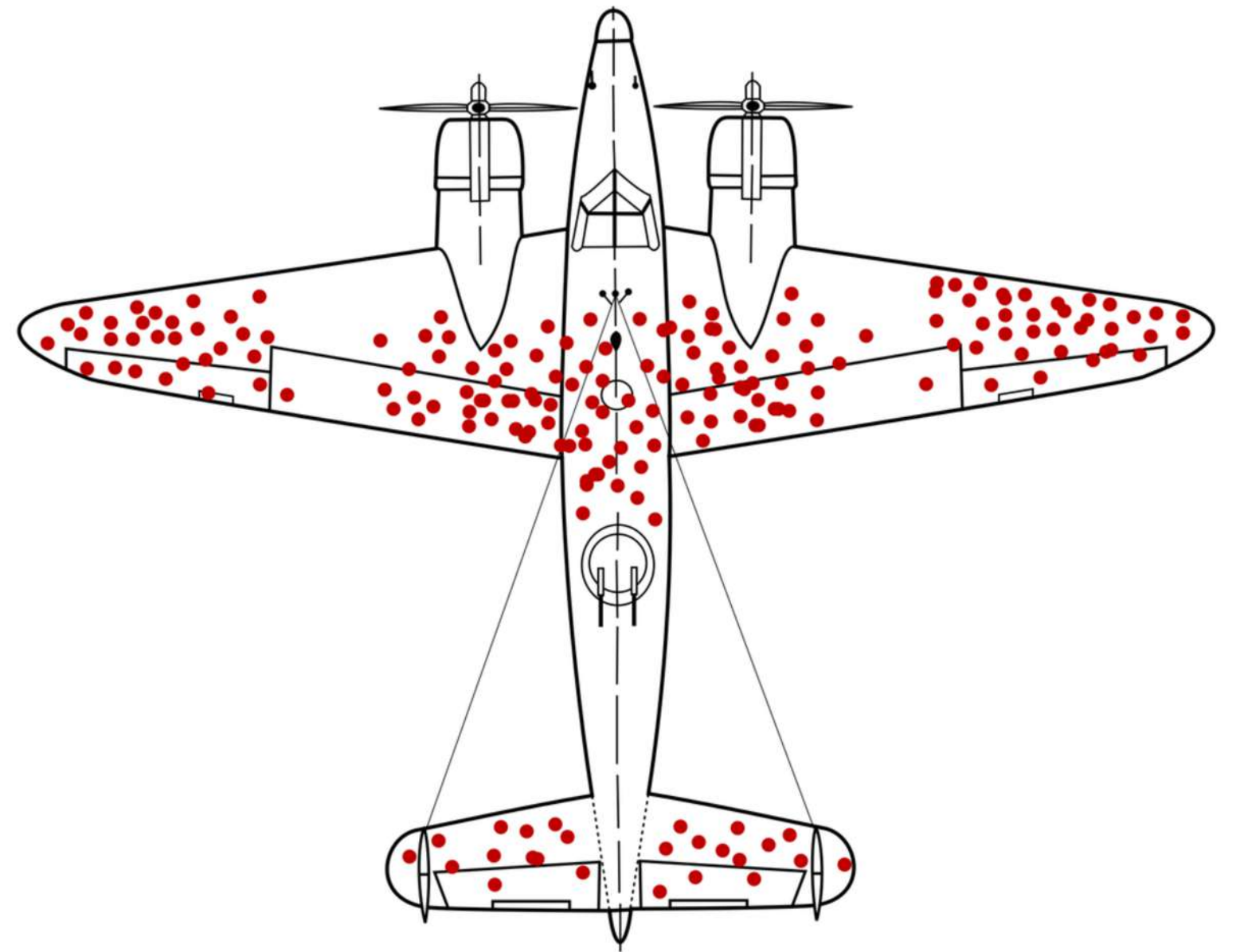
- However, statistician Abraham Wald pointed out a crucial problem: *the analysis included only planes that survived and returned.*



source: Wikipedia

About gaps in data: Survivorship Bias

- The missing data were the planes that had been shot down.
- Therefore, the areas with fewer bullet holes on surviving aircrafts were actually the most critical, because planes hit there did not survive.
- Drawing conclusions only from successful or surviving cases while ignoring the missing failures



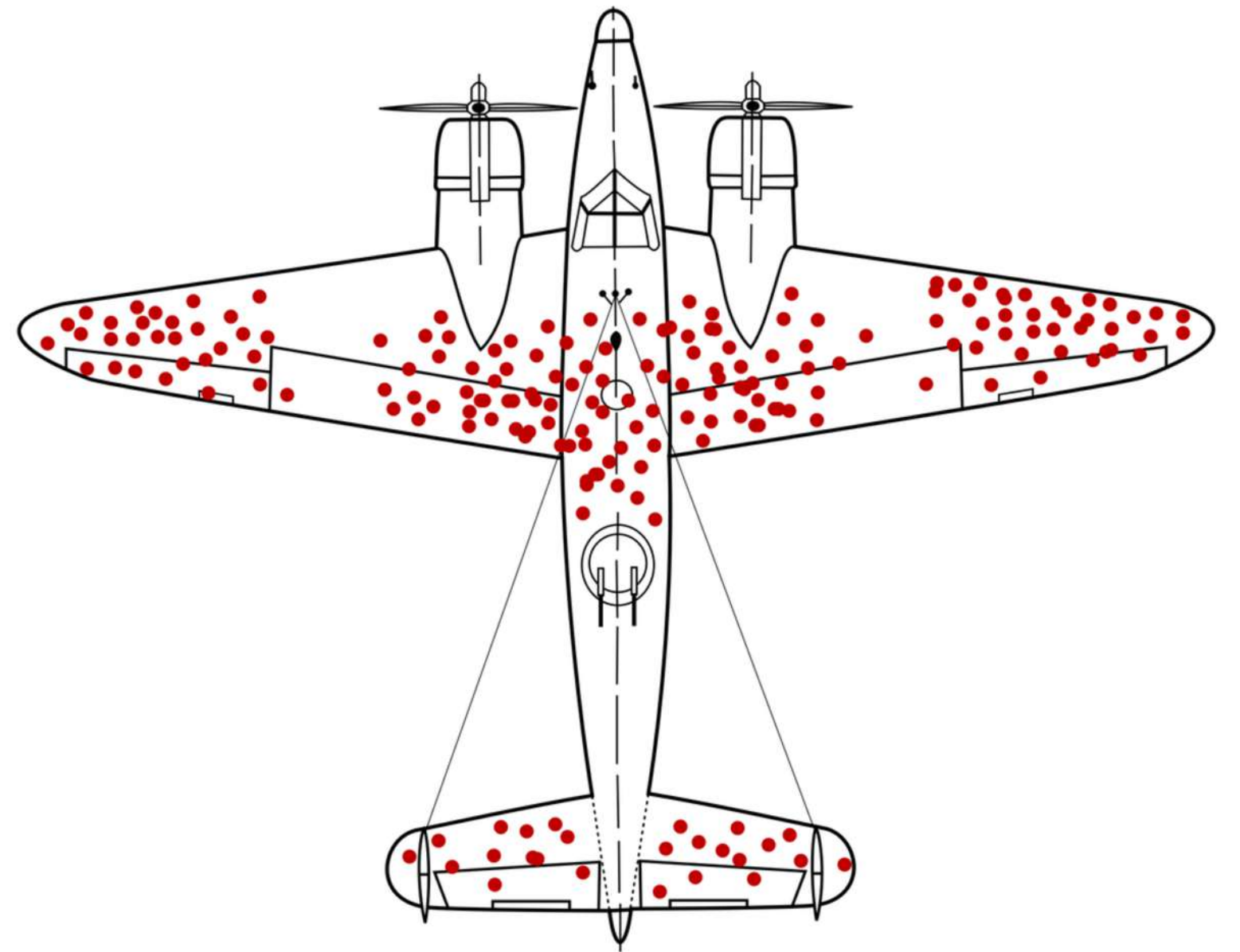
source: Wikipedia

About gaps in data: Survivorship Bias

Remember:

- Absence of evidence is not evidence of absence
- The missing data may matter
- We don't know what we don't know

Be careful when selecting data and drawing conclusions
from it.

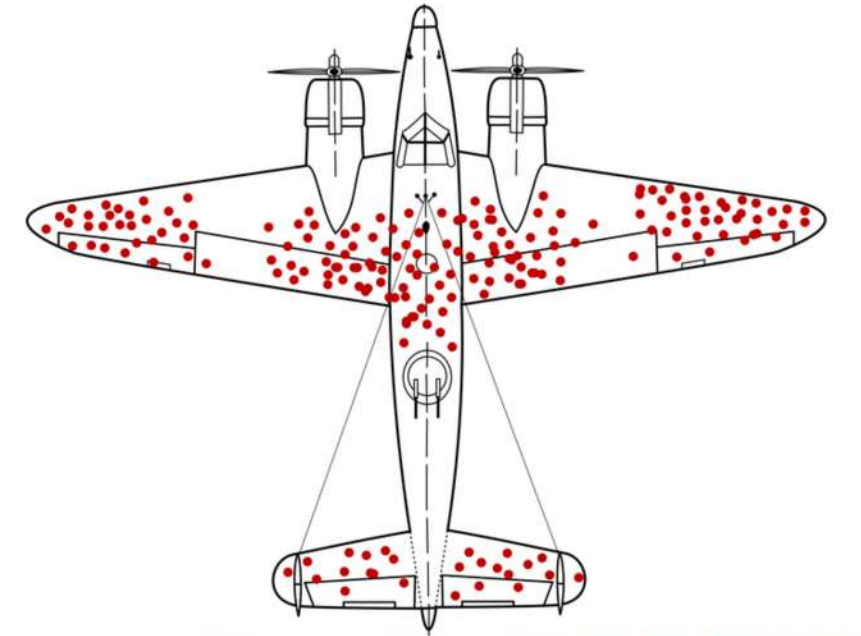


source: Wikipedia

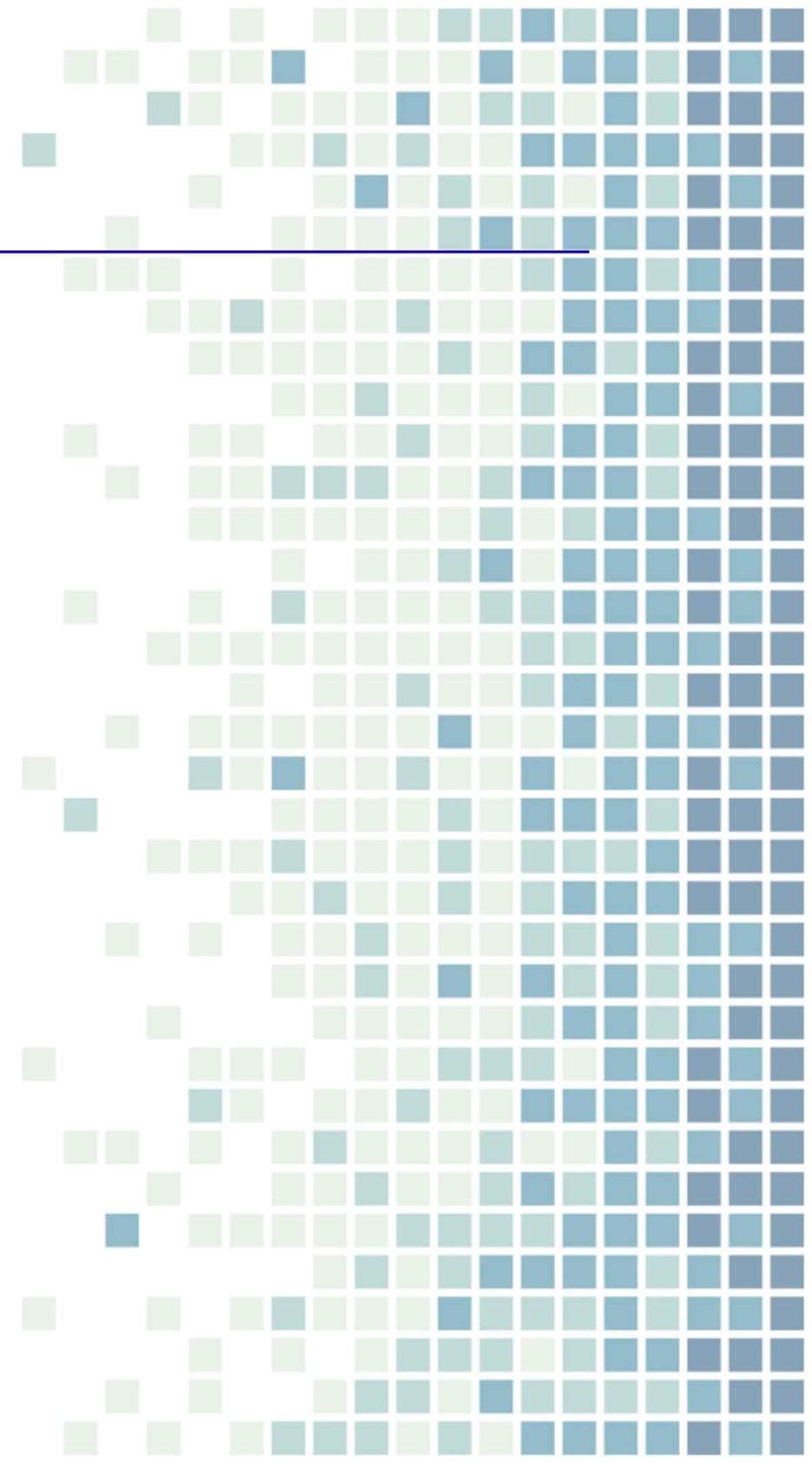
About gaps in data: Survivorship Bias

Read the full article:

https://en.wikipedia.org/wiki/Survivorship_bias

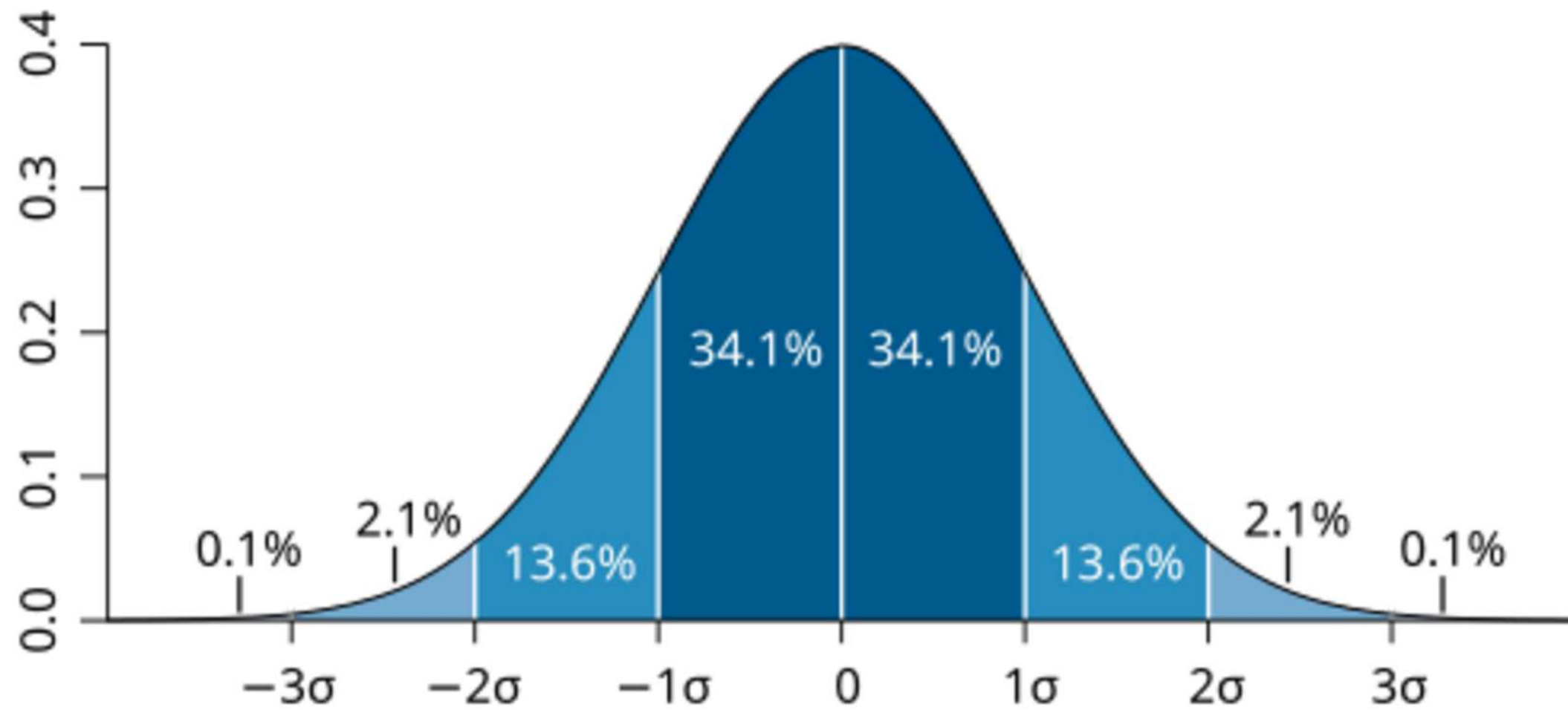


Gaps in data: Zipf's Law



Gaps in data: Zipf's Law

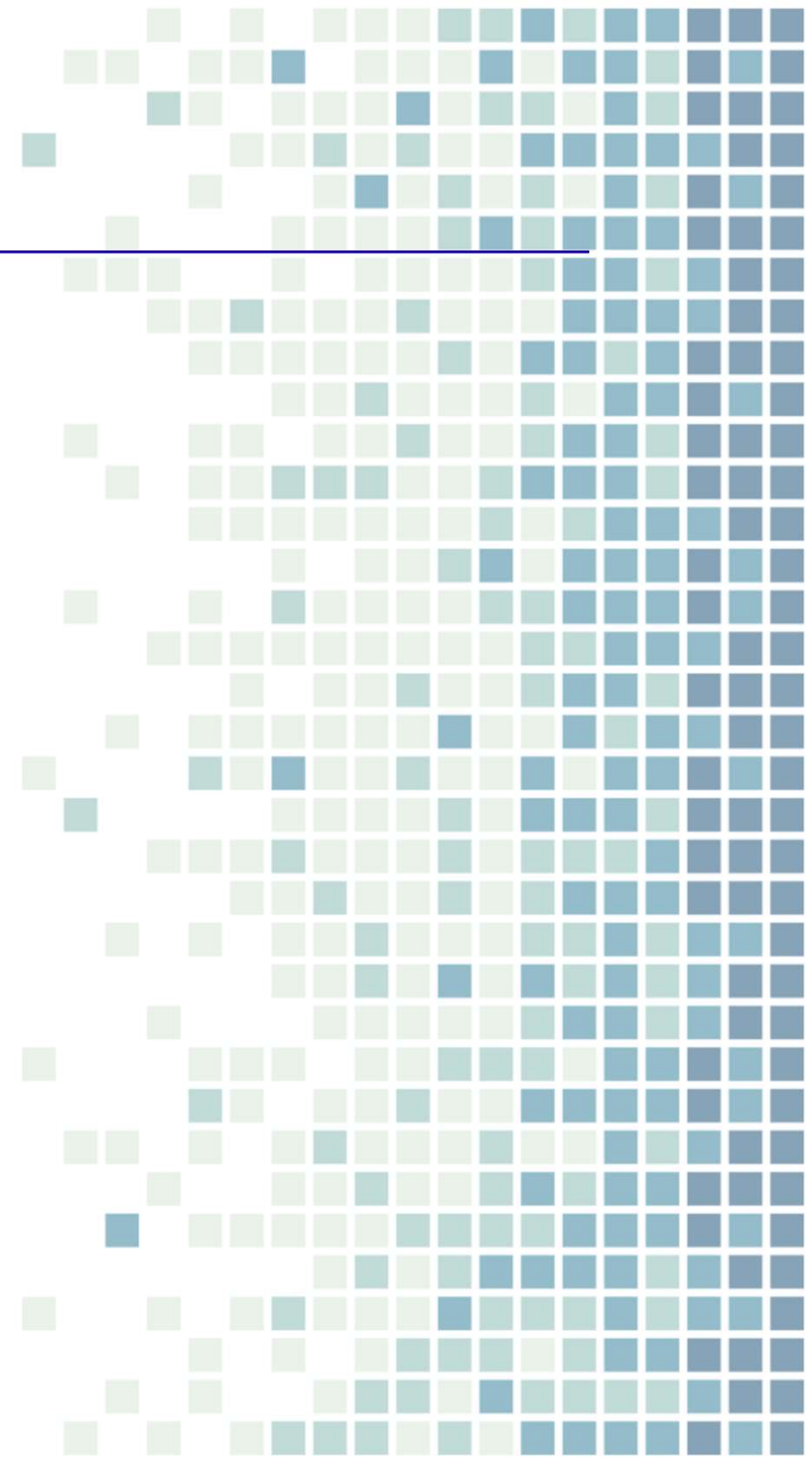
Normal (Gaussian) Distribution



Gaps in data: Zipf's Law

Normal (Gaussian) Distribution

- very common in the real world
- height, weight, IQ, measurement errors
- the most common values are close to the mean
- extreme values are rare
- symmetric



Gaps in data: Zipf's Law

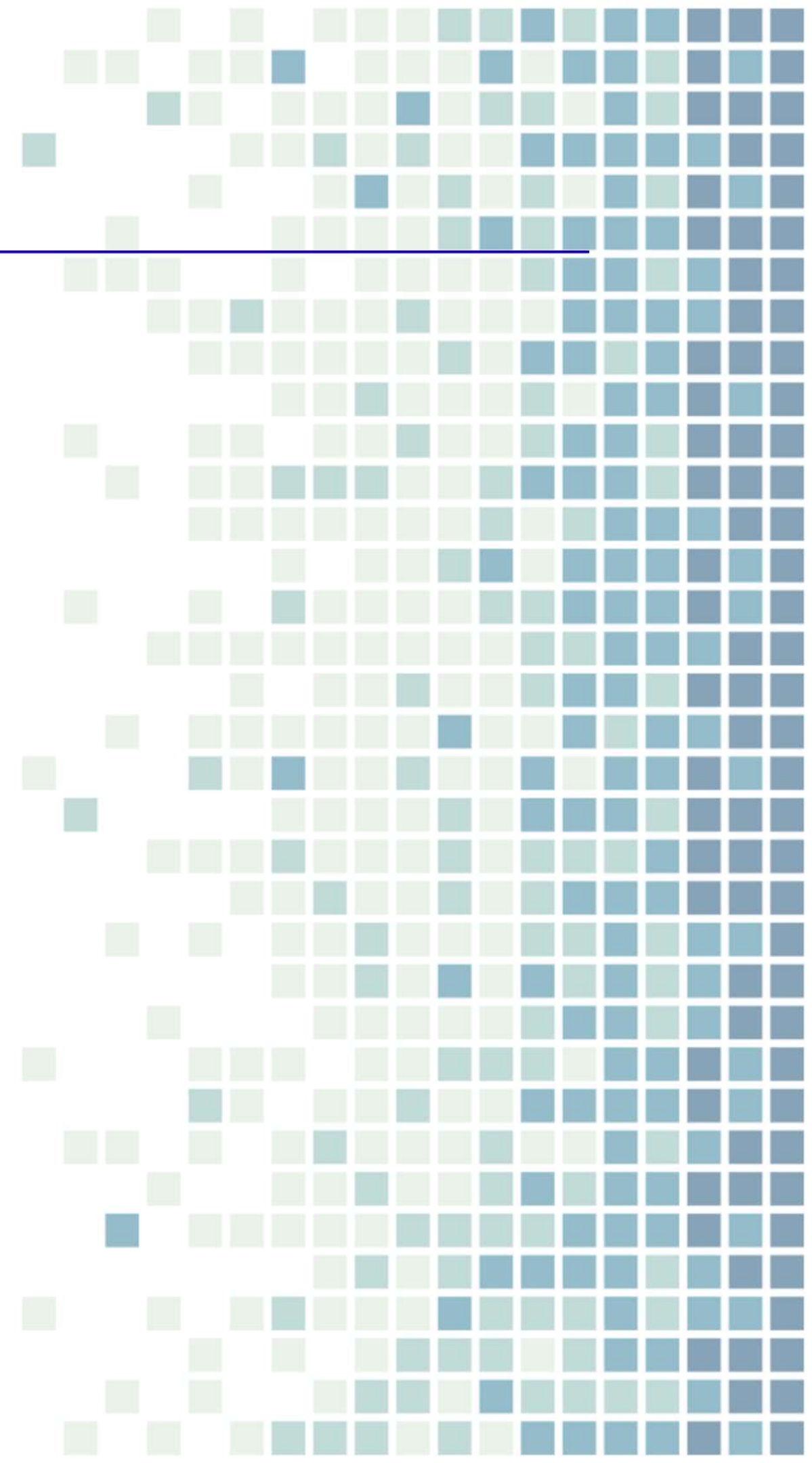
Power Law Distribution



Gaps in data: Zipf's Law

Power Law Distribution

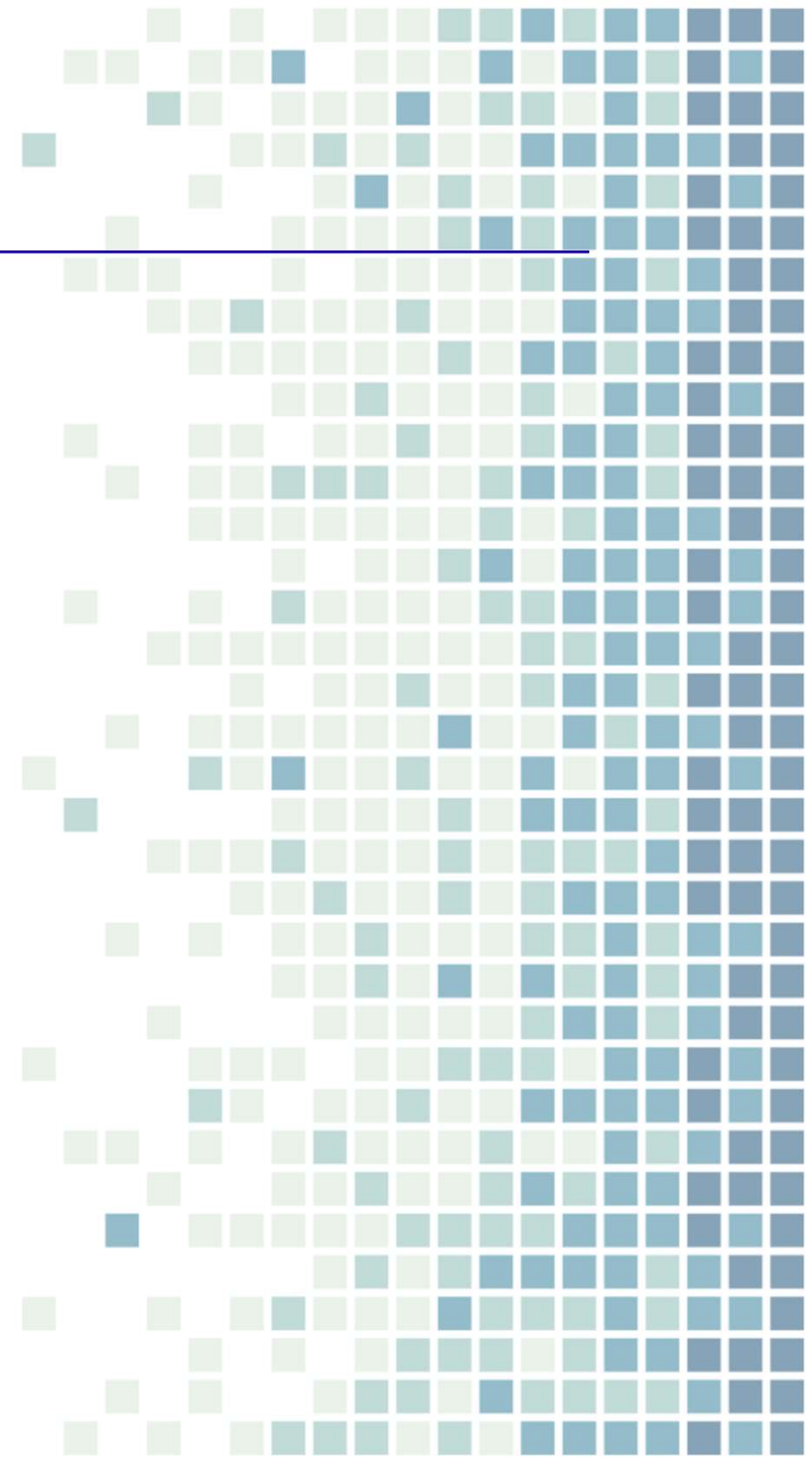
- less common; occurs in specific cases
- word frequencies, city sizes, wealth
- a strong dominance of the most frequent items
- the mean can be misleading
- asymmetric



Gaps in data: Zipf's Law

Zipf's Law

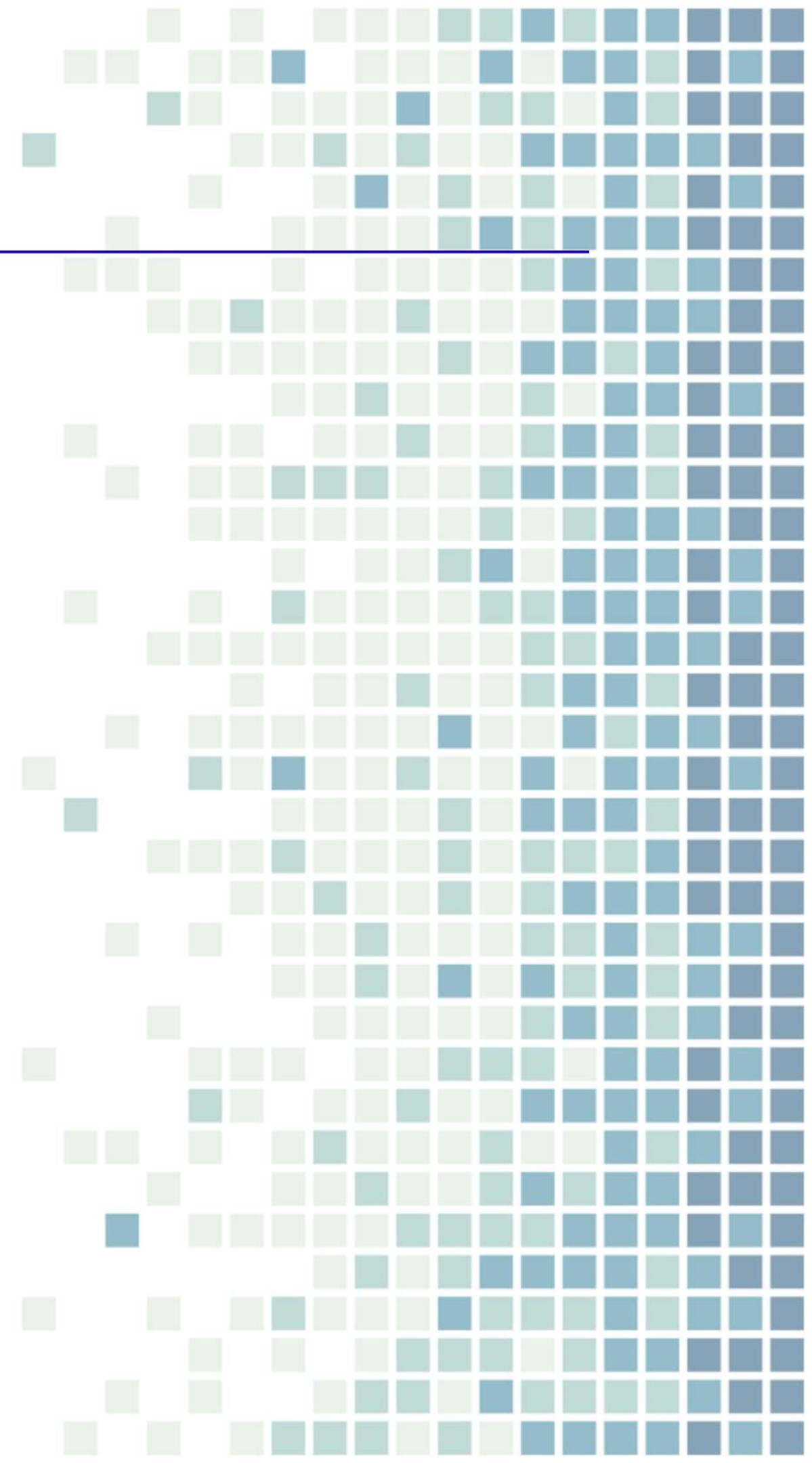
$$\text{word frequency} \propto \frac{1}{\text{word rank}} .$$



Gaps in data: Zipf's Law

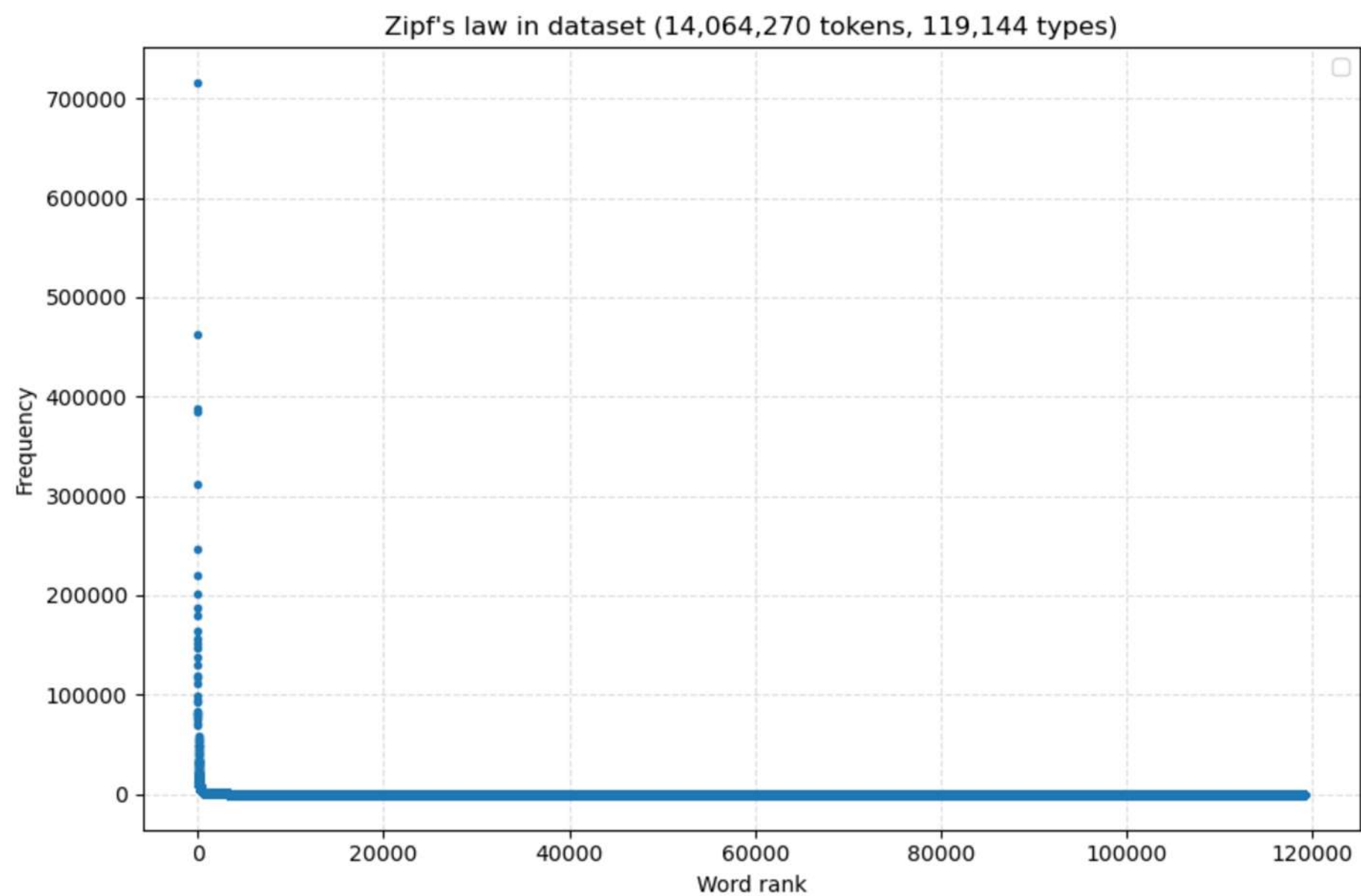
Zipf's Law

- a case of a power-law distribution
- connects word rank and frequency
- found in all natural languages



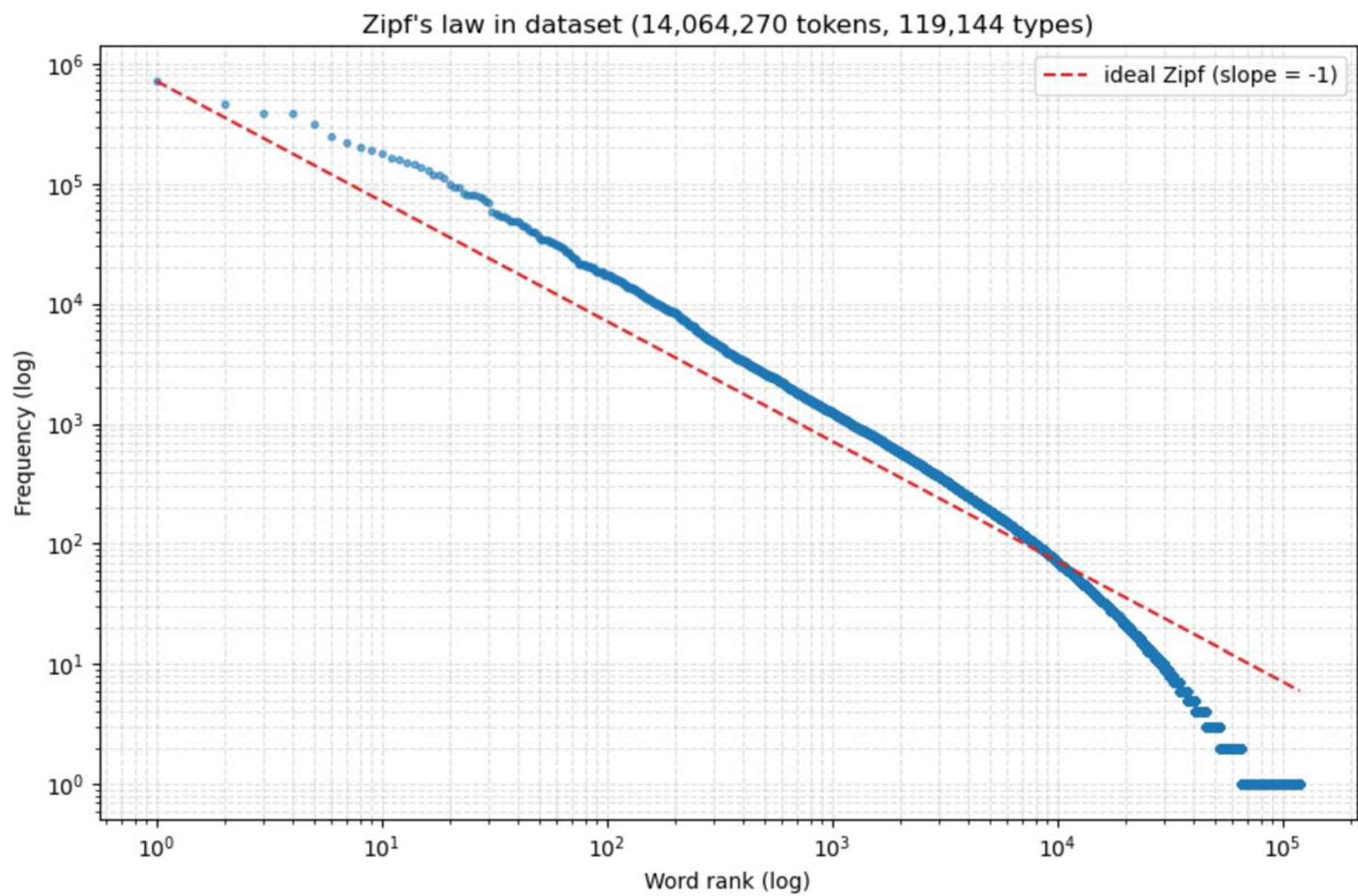
Gaps in data: Zipf's Law

Zipf's Law



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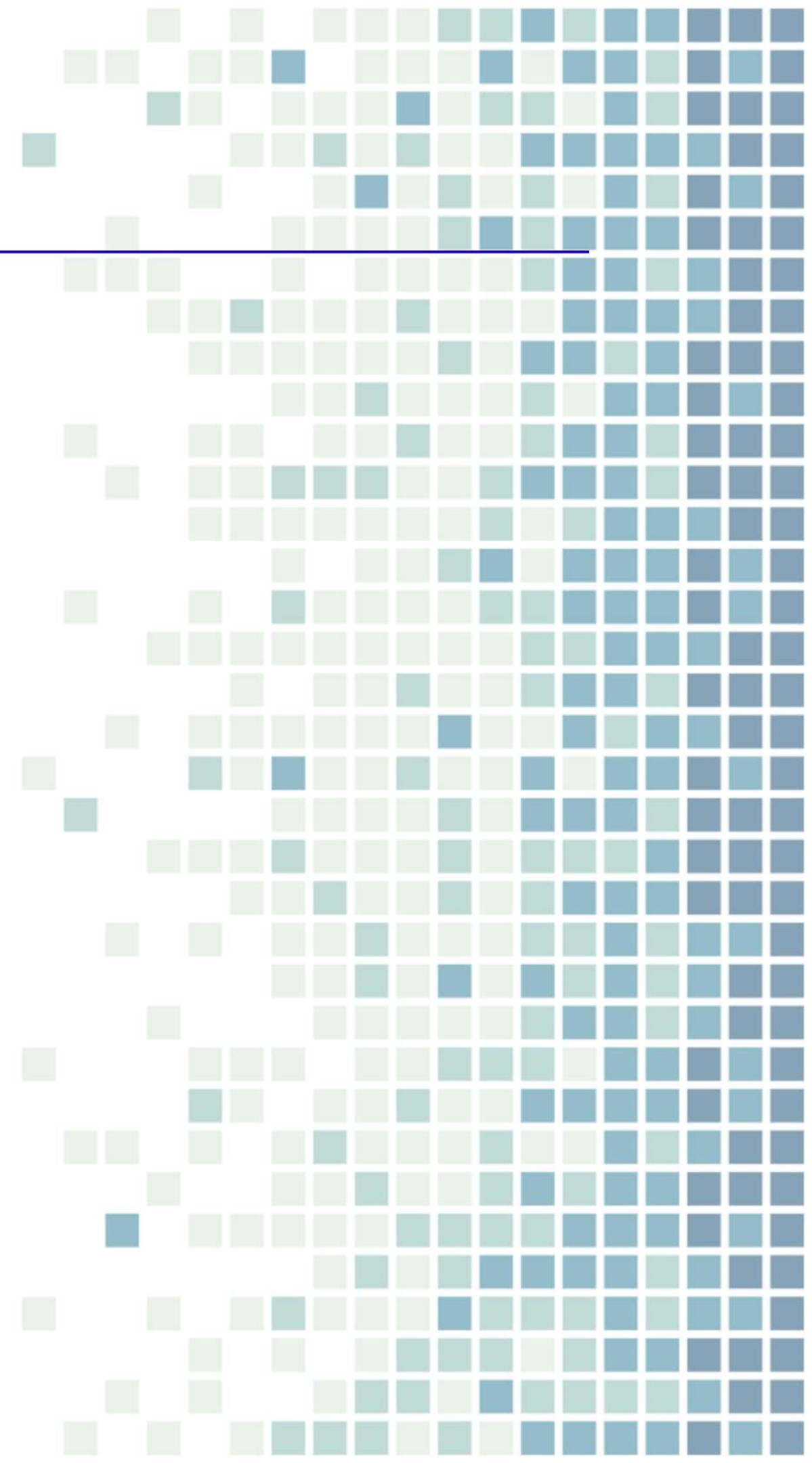
Zipf's Law



Gaps in data: Zipf's Law

Zipf's Law

- very long tail: many rare words
- even more 'zero-frequency' words



Gaps in data: Zipf's Law

Zipf's Law - interesting facts

- out of > 100 000 types (lexemes) in the 100 English Novels Corpus, the first make up half of all the tokens
- hapax legomena (words that occur only once) make up
 % of all types (/ 119144)

Gaps in data: Heaps's Law

HEAPS' LAW

$$V = k * M^{\beta}$$

V: Vocabulary Size
(Number of Unique words or 'Types' in the collection)

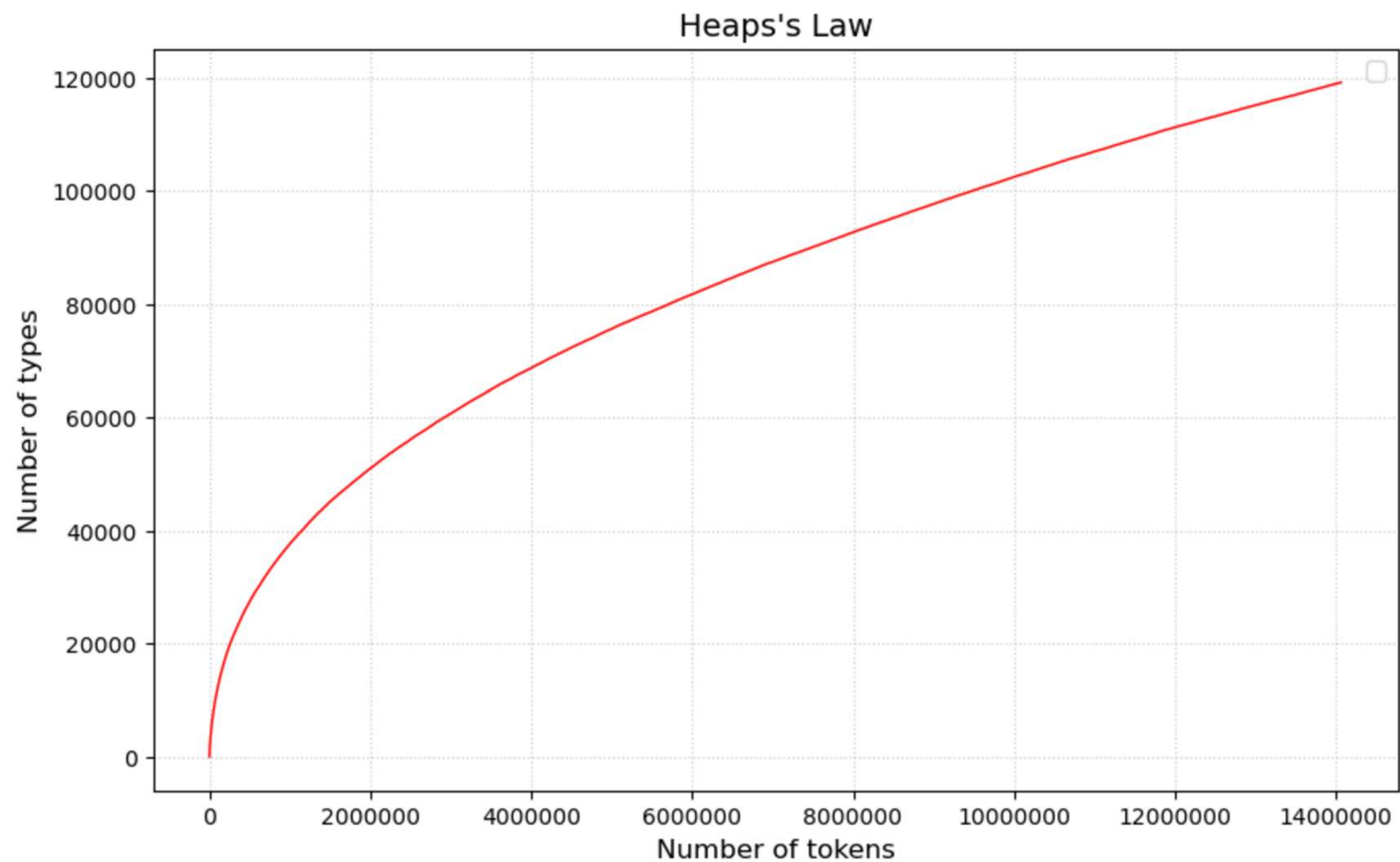
= : equals

k: Model Constant
(A coefficient related to language or text source; typically 10 to 100)

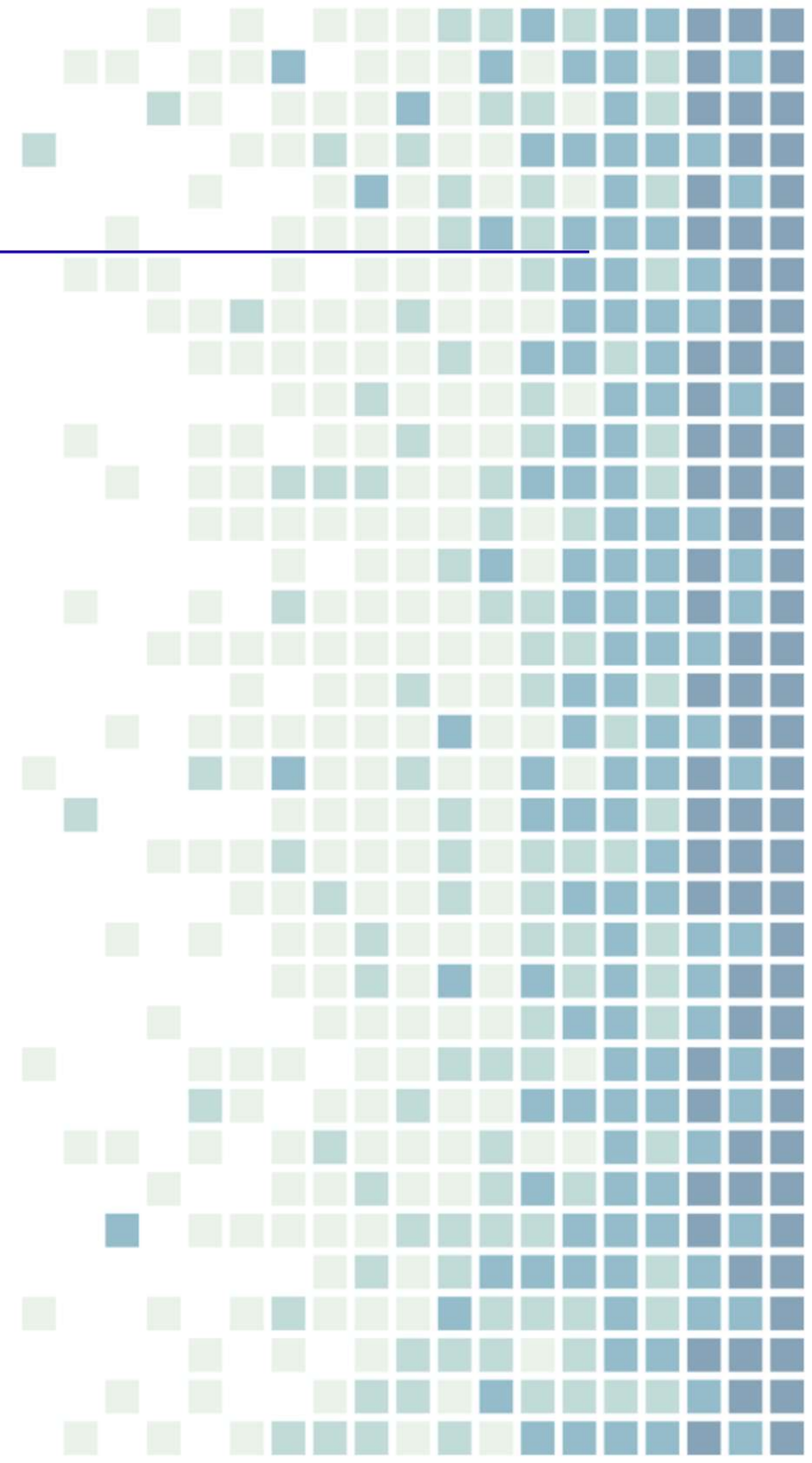
M: Number of Tokens
(Total number of words processed from the document or corpus)

β : Model Exponent
(A parameter determining curve steepness; usually between 0.4 and 0.7)

Gaps in data: Heaps's Law

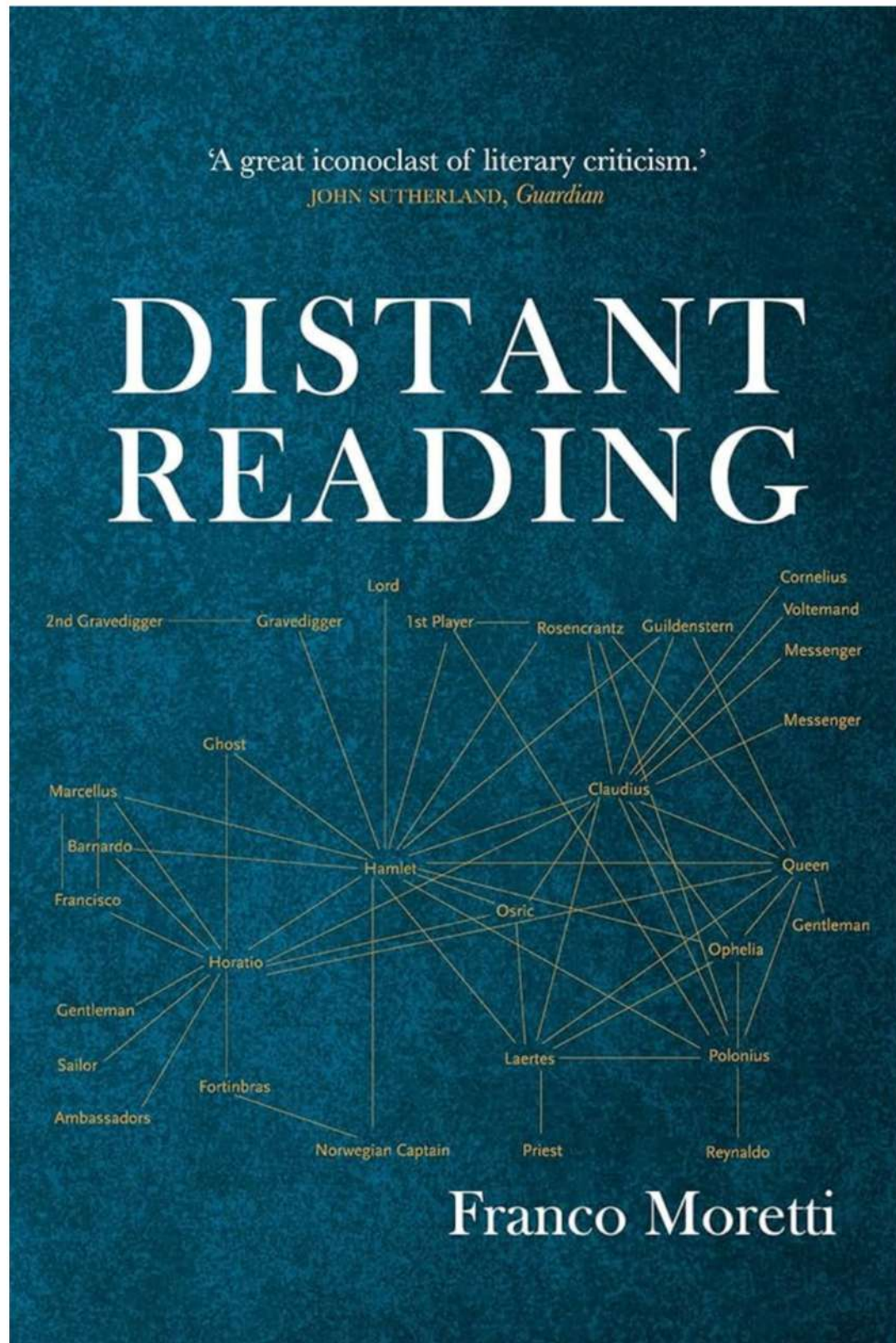


Gaps in data: Heaps's Law

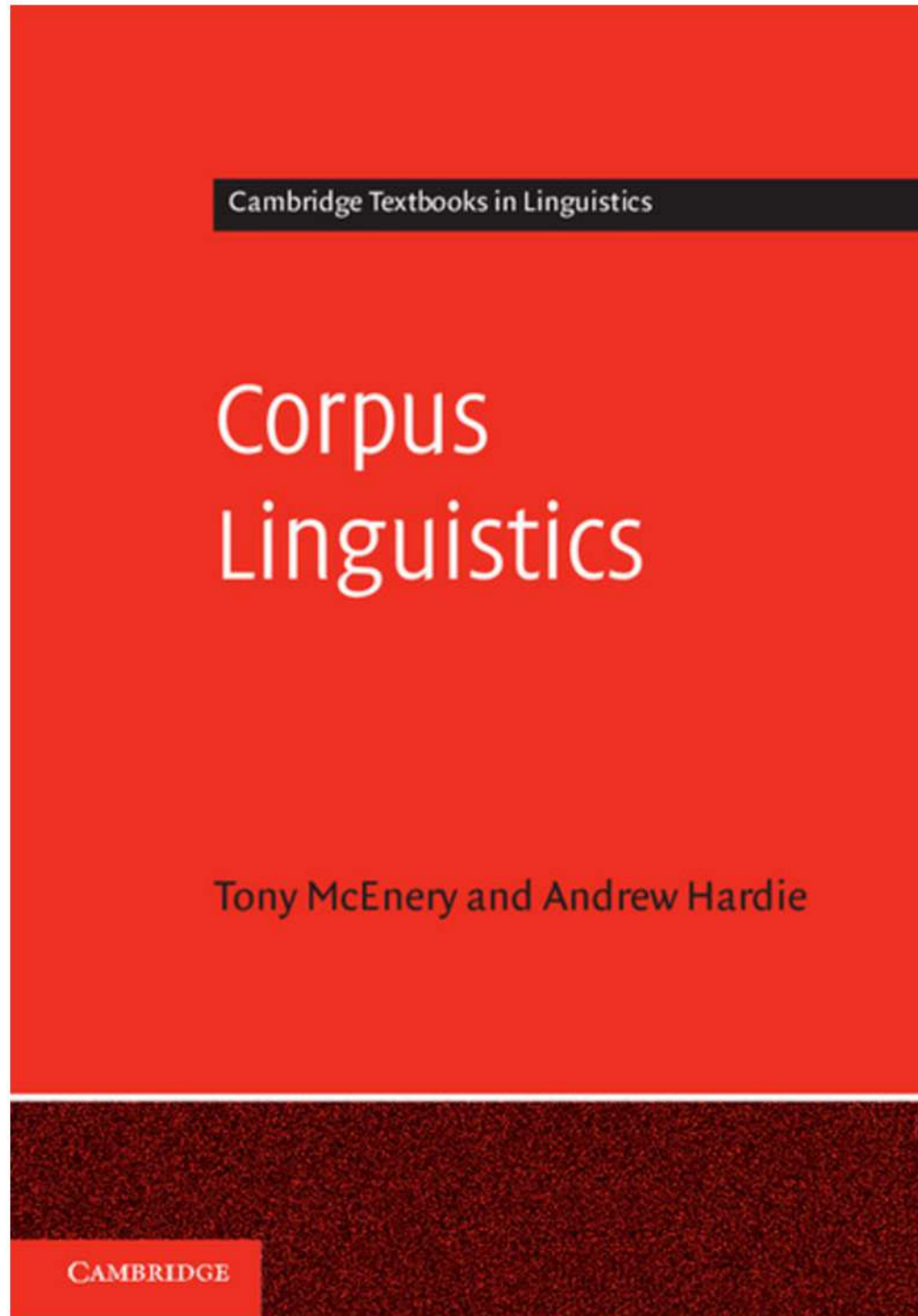


Further readings:

Franco Moretti: *Distant Reading*

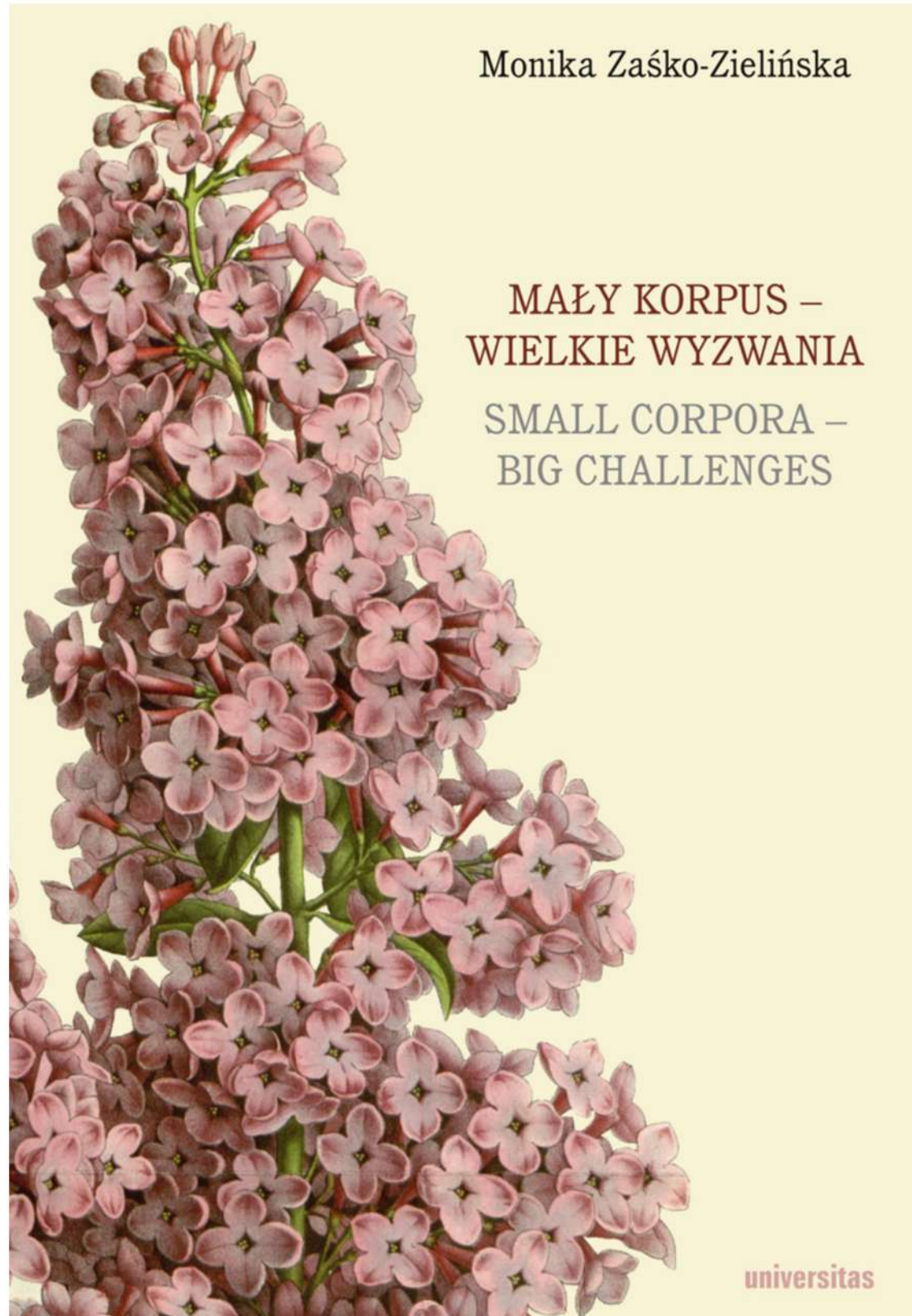


Further readings:



Tony McEnery and Andrew Hardie: Corpus Linguistics. Method, Theory and Practice

Further readings:



Monika Zaśko-Zielińska: *Small Corpora — Big Challenges*

[Link](#)